

KINGDOM OF SAUDI ARABIA

Ministry of Higher Education

Taibah University

College of Computer Science and

Engineering (Boys Section)



المملكة العربية السعودية
وزارة التعليم العالي

جامعة طيبة

كلية علوم وهندسة الحاسب الآلي

(قسم الطلاب)



Technical Support System

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Abstract

This graduation project aims to improve the technical support system at Taibah University, enhancing both efficiency and user satisfaction. This project identified that the current paper-based method for reporting technical issues creates significant delays and frustration among users. Our proposed solution is a computerized technical support system that streamlines the process of reporting, tracking, and managing these issues. By digitizing the support experience, users will benefit from real-time updates, greater transparency, and quicker responses. This system is designed to empower students, faculty, and technical staff, ensuring that technological challenges are addressed promptly and effectively, which will contribute to a better learning environment. Ultimately, this project not only tackles current problems but also lays the groundwork for ongoing improvements in service delivery at the university

Acknowledgment

Allah, the Creator of the Worlds, be praised. First of all, thanks be to God,

We would like to express our heartfelt thanks to everyone who supported us during the development of this graduation project. First and foremost, we are deeply grateful to our supervisor, Dr. Rashiq Marie, whose guidance and feedback were invaluable throughout this process. His expertise and encouragement helped shape our ideas and improve our work.

We also want to thank our fellow students and faculty members at Taibah University for their participation and insights. Their experiences provided us with a better understanding of the challenges related to technical support.

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Chapter 1: Introduction

1.1 Introduction

Technical support systems play a crucial role in organizations, especially within educational settings, where the integration of technology is essential for effective learning and administration. These systems are designed to provide assistance to users facing technological challenges, ensuring that both staff and students can maximize the benefits of digital tools. One of the primary goals of implementing robust technical support systems is to improve operational efficiency. By streamlining processes and reducing downtime, educational institutions can allocate more resources to teaching and learning, rather than troubleshooting technical issues.

In addition to enhancing operational efficiency, technical support systems significantly improve the customer experience. In educational environments, customers are not only the students but also faculty, staff, and parents who rely on technology for communication and information access. A responsive technical support system can lead to increased satisfaction among these groups, fostering a more conducive learning atmosphere. Furthermore, creating a comprehensive database for common issues allows organizations to track recurring problems and develop proactive solutions. This database serves as a valuable resource for both users and support staff, facilitating quicker resolutions and reducing the burden on support teams.

With the rapid pace of technological advancement, educational institutions face the challenge of keeping up with new tools and platforms. As technology evolves, so do the complexities of its use. Effective technical support systems are essential for responding to these changes, enabling users to navigate new technologies confidently. This responsiveness not only addresses immediate concerns but also fosters an environment of continuous learning and adaptability, which is vital in today's fast-paced educational landscape. By prioritizing technical support, organizations can ensure that they are well-equipped to meet the diverse needs of their community.

1.2 Problem Definition

At Taibah University, the management of technical faults within computer labs poses significant challenges that hinder operational efficiency and negatively impact students' educational experiences. One of the primary issues is the reliance on a paper-based reporting system for faults and repairs. This method introduces a number of inefficiencies that complicate the fault management process.

Firstly, the lack of a centralized tracking system for repairs leads to difficulties in monitoring the status of reported issues. When students or staff submit a fault report on paper, there is often no immediate acknowledgment or tracking mechanism to ensure that the issue is being addressed. As a result, repairs can be delayed or overlooked entirely, leaving students without access to essential technologies for extended periods. The absence of real-time updates creates frustration and dissatisfaction among users who are left in the dark regarding the resolution of their issues.

Additionally, the paper-based system increases the burden on technical staff, who must manually sort through numerous reports, prioritize them, and follow up with repairs. This not only consumes valuable time but also diverts their attention from proactive maintenance and improvements in the lab environment. The inefficiency of this process can lead to technical staff experiencing burnout, further exacerbating the challenges faced in managing the computer labs.

The impact on students is profound. Extended downtimes of critical equipment can disrupt learning activities, diminish the quality of educational experiences, and hinder students' ability to complete assignments efficiently. When technology fails, the ripple effects extend beyond the labs, affecting overall academic performance and satisfaction. Consequently, the current paper-based fault management system at Taibah University is in dire need of re-evaluation and modernization to better serve the needs of its students and staff.

1.3 Project Objectives

The primary objective of establishing a computerized technical support system at Taibah University is to significantly improve service quality across all user interactions. This entails creating a streamlined process for logging and tracking technical issues, thereby ensuring that all reported faults receive prompt attention. By digitizing the reporting system, users will have the ability to see real-time updates on the status of their requests, fostering a sense of reliability and trust in the technical support process.

Enhancing operational efficiency stands as another critical goal of this project. By implementing a computerized system, this project aims to reduce the time spent by technical staff on manual reporting and follow-ups. Automating these tasks will enable technicians to focus on resolving issues rather than managing paperwork. Additionally, the system will facilitate better data management, allowing for the analysis of trends in technical faults, which can inform future preventive measures and resource allocation.

Accelerating fault repairs is also a key objective. The new system will prioritize issues based on severity and impact, allowing technicians to address the most critical problems first. By utilizing a tiered response system, this project can ensure that urgent matters are handled swiftly, minimizing downtime for students and staff who rely on technology for their educational activities.

Another important aspect of the project is increasing transparency for users. With a computerized system, users will have access to detailed information regarding the status of their reported issues, including expected resolution times. This transparency will empower users by keeping them informed and engaged in the support process.

Finally, optimizing technician allocation is essential for maximizing the effectiveness of the technical support team. By analyzing patterns in fault reports, the system can help identify peak times for technical issues and allocate technicians accordingly. This strategic deployment of resources will not only enhance response times but also ensure that technical staff are utilized to their fullest potential, thereby improving overall service delivery.

1.4 System Scope

The proposed technical support system at Taibah University aims to create a comprehensive framework that caters to various categories of users, devices, and processes, ensuring a robust response to technical challenges encountered within the educational environment.

The system will serve multiple user categories, which include:

- **Students:** These individuals will primarily report issues related to their learning tools, such as computers and projectors, requiring timely support to facilitate their academic activities.
- **Faculty:** Faculty members will utilize the system to address issues with teaching aids and classroom technology, ensuring that their instructional methods remain uninterrupted.
- **Technicians:** Technical staff will be responsible for managing and resolving reported issues, utilizing the system to track and prioritize tasks efficiently.
- **Supervisors:** Supervisors will oversee the technical support operations, leveraging data from the system to monitor performance, resource allocation, and user satisfaction.

The technical support system will streamline several key processes:

- **Reporting Issues:** Users will be able to submit issues electronically, ensuring that all reports are logged and acknowledged promptly.
- **Task Assignment:** The system will automate the assignment of tasks to technicians based on availability and expertise, enhancing efficiency.
- **Status Tracking:** Users will have access to real-time status updates on their reported issues, improving transparency and communication.
- **Report Closure:** Once an issue is resolved, the system will facilitate the closure of reports, allowing users to confirm resolution and providing feedback on their experience.

Implementation Location and Future Expansion

The initial implementation of the technical support system will take place within the computer labs of Taibah University, where the need for efficient fault management is most pressing. Furthermore, the system is designed with future expansion in mind, allowing for integration with additional departments and facilities as the university's technological needs evolve. This adaptability will ensure that the technical support system remains relevant and effective, providing a foundation for ongoing improvements in service delivery.

1.5 Project Timeline

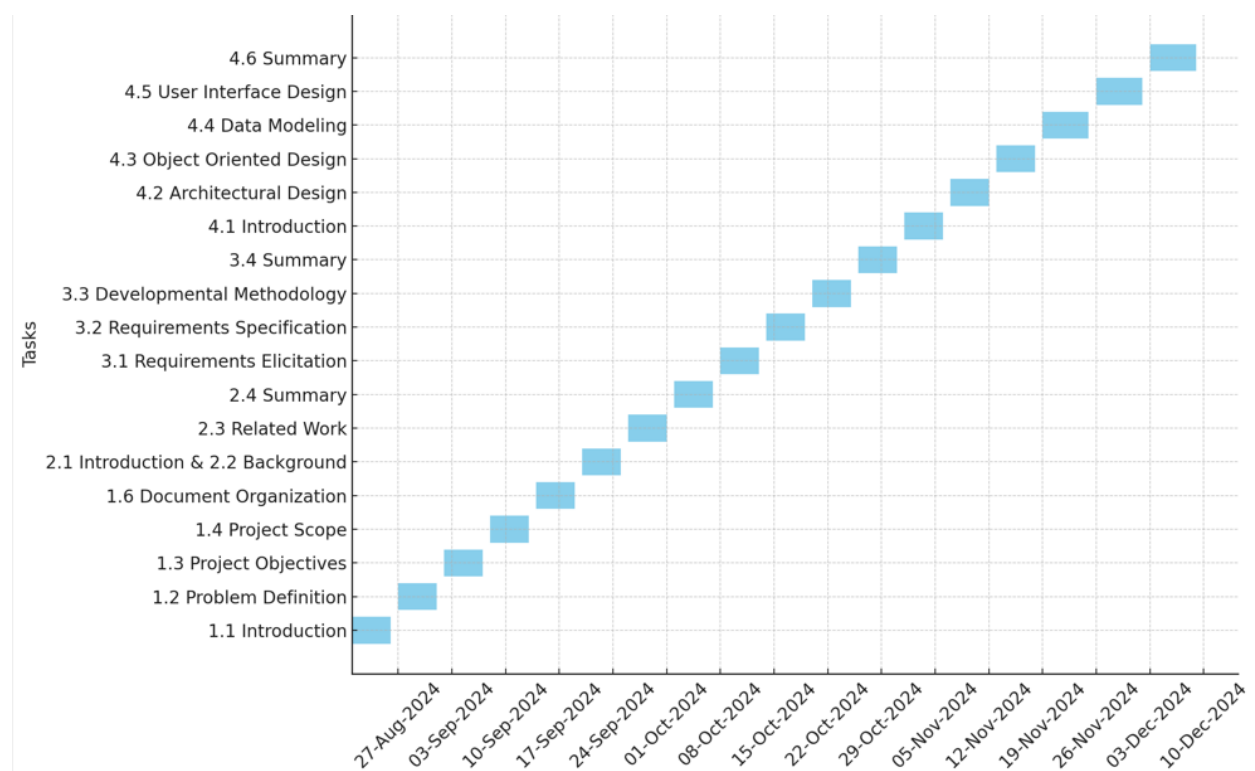


Figure 1.5.1 Gantt Chart

1.6 Document Organization

Chapter 1: Introduces the project objectives, defines the problem, presents the Gantt chart, outlines the scope of work, and provides an overview of the project layout.

Chapter 2: Discusses the literature review, background information, and the relationship of relevant works to our project, highlighting how these connections inform our approach.

Chapter 3: Introduces the system analysis, detailing both functional and non-functional requirements, user requirements, and the methodology applied in the project.

Chapter 4: Represents the system design, including structural static models (such as class diagrams and ERDs), dynamic models, and interface designs.

1.7 summary

In this chapter, this project examines the problem, goals, scope, and timeline. It provides a brief summary of the upcoming chapters, which will set the stage for the analysis and design phases. Before moving forward, this project will explore related works to gain further insights that will inform its development.

Chapter 2 Literature Review

2.1 Introduction

In this chapter, this project will delve into the critical importance of implementing a robust technical support system within educational institutions. The primary aim is to establish a seamless framework that not only addresses technical issues but also enhances overall staff productivity. A well-structured technical support system serves as the backbone of effective educational operations, allowing educators and administrative staff to focus on their core responsibilities rather than being bogged down by unresolved technological challenges.

The resolution of technical issues is paramount, as it directly impacts the ability of staff to perform their duties efficiently. When technology functions smoothly, educators can deliver lessons without interruption, and administrative personnel can manage their tasks effectively. This improved productivity is essential in fostering a positive educational environment where both students and faculty can thrive. By mitigating the frustrations associated with technical malfunctions, this project create a culture of reliability and support that encourages engagement and active participation from all stakeholders.

Furthermore, a responsive technical support system contributes to an enriched educational experience for students. In today's technology-driven learning landscape, students rely heavily on various digital tools for their studies. When these tools encounter problems, it can lead to significant disruptions in their learning processes, resulting in decreased satisfaction and academic performance. Thus, swift and effective issue resolution is not just an operational necessity; it is a vital component of ensuring that students receive the quality education they deserve.

As this project explore the implementation of this system, this project will address specific strategies and objectives aimed at transforming the current state of technical support. By prioritizing user needs and enhancing communication between technical staff and users, this project can build a foundation for a more effective and engaging educational environment.

2.2 Background

In today's educational landscape, effective technical support has become a necessity rather than a luxury. As students and faculty increasingly rely on technology for learning and teaching, the ability to resolve technical issues promptly is crucial. A typical student experience in a computer lab, however, often reveals the inadequacies of current support systems. When faced with a malfunctioning computer or software, students may resort to a cumbersome manual reporting process. They fill out paper forms, often without any immediate feedback or assurance that their issue is being addressed. This lack of transparency not only frustrates students but also hinders their academic performance.

The drawbacks of existing manual reporting systems are manifold. First, it creates a disjointed communication channel between students and technical staff. Without a centralized system to track reported issues, students are left uncertain about the status of their requests. This uncertainty breeds anxiety, especially when deadlines loom and access to technology is critical for completing assignments. Additionally, the manual process places undue stress on technical staff, who must sift through stacks of paper reports and prioritize them based on urgency. This can lead to burnout and decreased morale among support personnel, ultimately affecting the quality of service provided.

Advocating for a shift toward a computerized system is vital for addressing these challenges. A digital platform would streamline the reporting process, allowing students to submit issues electronically and receive real-time updates on their status. This not only enhances communication but also significantly reduces the administrative burden on technical staff. By automating task assignments and tracking repairs, technical teams can focus on resolving issues more efficiently, leading to faster response times and improved user satisfaction. In essence, transitioning to a computerized technical support system is not just an upgrade; it is an essential step toward creating a more responsive and effective educational environment.

1- Taibah university system

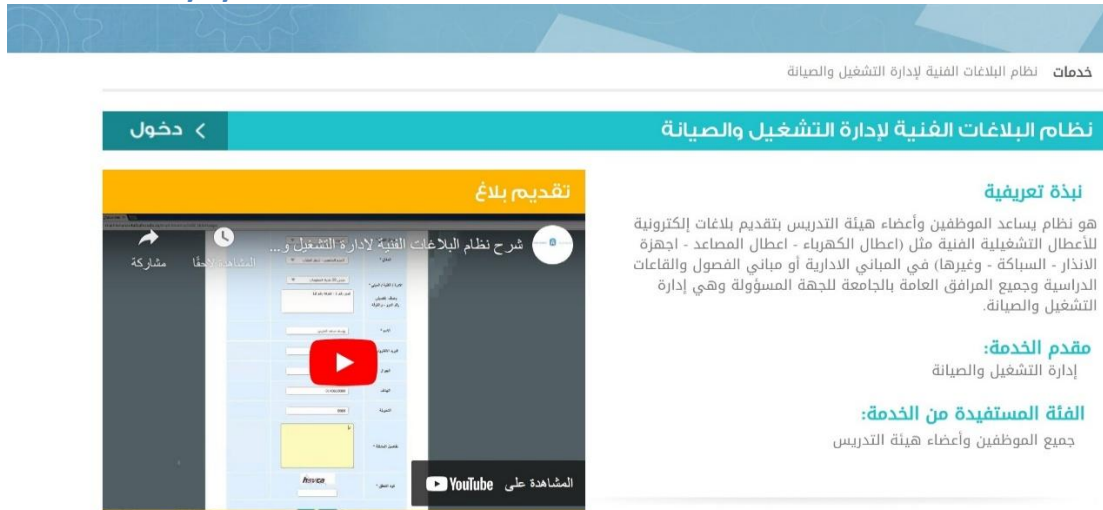


Figure 2.3.1 Taibah university system



Figure 2.3.2 Taibah university system

The system is very simple; it consists of interfaces that can be accessed via the university's website by selecting the issue and providing some details about the request, such as email, name, and information about the problem, including the reason for the issue, the building number, and the room where the problem occurred. The report is sent to the top management (supervisor), who then assigns the report to the relevant department. The department assigns the issue to a technician to address the case. The handling of the case depends on the commitment of the employee, as there are some less critical reports that can be easily resolved without involving technical support staff

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KING ABDULAZIZ UNIVERSITY

عن الجامعة | القبول | الأكاديمية | البحث والإبتكار | الحياة الجامعية | الخدمات الإلكترونية

جامعة الملك عبدالعزيز

نظام (مرافق) يرفع رضا المستفيدين في التعامل مع بلاغات الصيانة والخدمات

[illegible]

The **Maraafiq system** integrates a streamlined reporting process that allows users to submit technical issues electronically. This system provides real-time tracking of reported problems, ensuring that users remain informed about the status of their requests. By automating the task assignment to technicians based on priority and availability, Maraafiq enhances the efficiency of the technical support team. The impact of this system is evident in the reduction of response times and improved user satisfaction, as students and faculty receive timely assistance with their technical issues.

3- Imam university support system

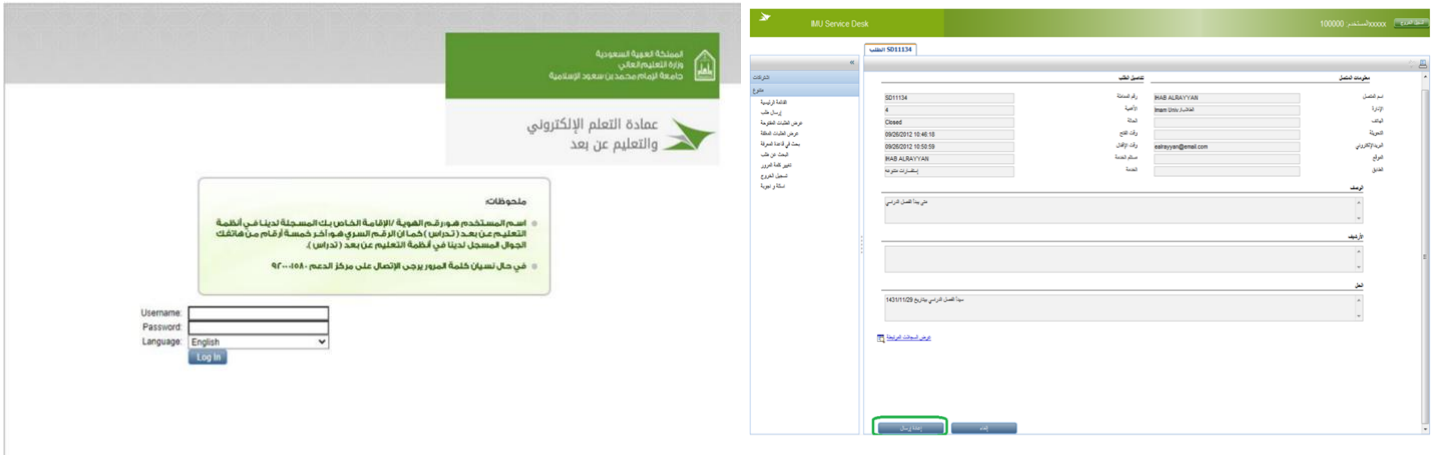


Figure 2.3.5 Imam university support system

Similarly, the **Imam University Support System** has gained recognition for its robust features. This system offers a user-friendly interface that simplifies issue reporting and tracking, while also providing a comprehensive database of frequently encountered problems. Such a repository enables both users and technical staff to reference common solutions, fostering a proactive approach to problem-solving. The effectiveness of the Imam University Support System is reflected in its ability to facilitate quicker resolutions and enhance overall service delivery.

4- King Khalid university system

المشرف العام	مساعد المشرف	رئيس القسم	التصعيد إلى
بعد مرور (2) ساعة، ولم يتم إغلاق البلاغ	بعد مرور (1) ساعة، ولم يتم إغلاق البلاغ	عند إحالة البلاغ للقمص	البلاغ الحرج
بعد مرور (3) ساعة، ولم يتم إغلاق البلاغ	بعد مرور (2) ساعة، ولم يتم إغلاق البلاغ	بعد مرور (1) ساعة، ولم يتم إغلاق البلاغ	البلاغ العالي
بعد مرور (٧٢) ساعة، ولم يتم إغلاق البلاغ	بعد مرور (٤٨) ساعة، ولم يتم إغلاق البلاغ	بعد مرور (٢٤) ساعة، ولم يتم إغلاق البلاغ	البلاغ المتوسط

Figure 2.3.6 King Khalid university system

رقم البلاغ	اسم صاحب البلاغ	تاريخ البلاغ	حالة البلاغ	نوع الخطب	الخطب
	الشيخ علي الجليل		قيد الانتظار - القسم الثاني		

عدد الخطبات 1

البلاغ	حركات البلاغ
رقم صاحب البلاغ:	اسم صاحب البلاغ:
جهة الموظف:	قسم الموظف:
حوالي الموظف:	توصية الموظف:
التبريد الإلكتروني:	رقم البلاغ:
تاريخ البلاغ:	حالة البلاغ:
جهة البلاغ:	قسم البلاغ:
المقر:	موظف قسم البلاغ:
نوع الخطب:	الخطب:
أولية البلاغ:	
ملاحظات البلاغ:	
ملاحظات القسم:	

Figure 2.3.7 King Khaled university system

King Khalid University's IT Reporting System uses a tiered response mechanism to prioritize issues by severity, ensuring critical problems are resolved quickly to minimize downtime. This efficient approach boosts operational reliability and user trust. The system demonstrates how universities can enhance technical support through features like real-time tracking, automated assignments, and comprehensive databases. These improvements serve as a valuable model for Taibah University's modernization efforts

2.4 Review Relevant Work

	<i>King khalid university</i>	<i>Imam mohammad ibn saud University</i>	<i>Taibah University</i>	<i>Maraafiq</i>
<i>Can a student submit a request?</i>	No	Yes	No	Yes
<i>Does it send directly to the support staff ?</i>	Yes	Yes	Yes	Yes
<i>Evaluate the degree of the problem</i>	No	No	No	Yes
<i>Proof of repair by report</i>	Yes	No	No	Yes
<i>Can the supervisor access the requests?</i>	Yes	Yes	Yes	Yes
<i>Is there a tracking system for submitted requests?</i>	No	No	No	Yes
<i>Can track the status of their requests ?</i>	Yes	Yes	No	Yes

Table 2.4 review relevant work

2.5 Summery

This chapter focuses on the need for a strong technical support system in Universities to help staff work better and solve tech issues quickly. It stresses that when technology works well, both faculty members and students are happier and perform better. The chapter points out problems with current paper-based reporting systems, which create communication issues and stress for staff. It also looks at existing systems like Maraafiq and Imam University Support System, showing how their useful features can help improve technical support in universities.

Chapter 3: System Analysis

3.1 Introduction

In this chapter, it will discuss the analysis and clarify the objectives outlined in the previous chapters. This chapter, this project examines the problem, goals, scope, and timeline. It provides a brief summary of the upcoming chapters, which will set the stage for the analysis and design phases. Before moving forward, this project will explore related works to gain further insights that will inform its development.

use diagrams to simplify the understanding of the system. This chapter includes comprehensive specifications of the requirements and functionalities needed to complete the system, along with the analysis process and the selected system model.

3.2 Analysis of Existing System

this section, dedicated to knowing and collecting the opinions of the proposed, and how they will accept the new web-site that will build to serve them in proper way. A questionnaire is designed and distributed to those users, 40 individual answers to the questionnaire and it were collected for their feedback, the results showed up as:

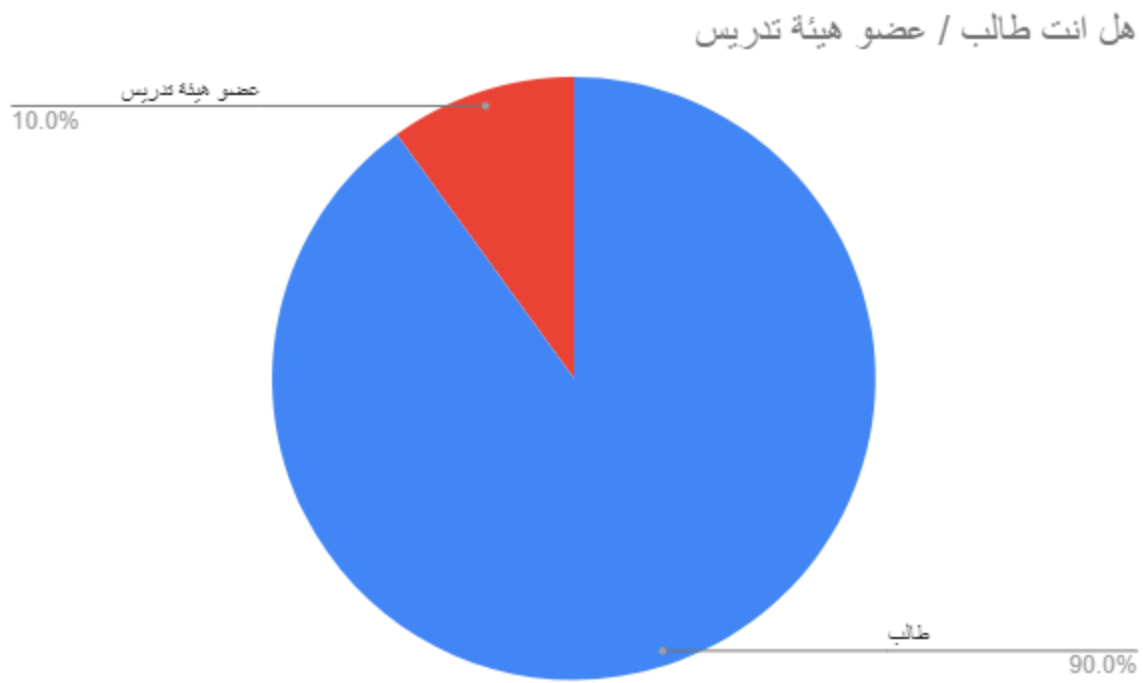


Figure 3.2.1

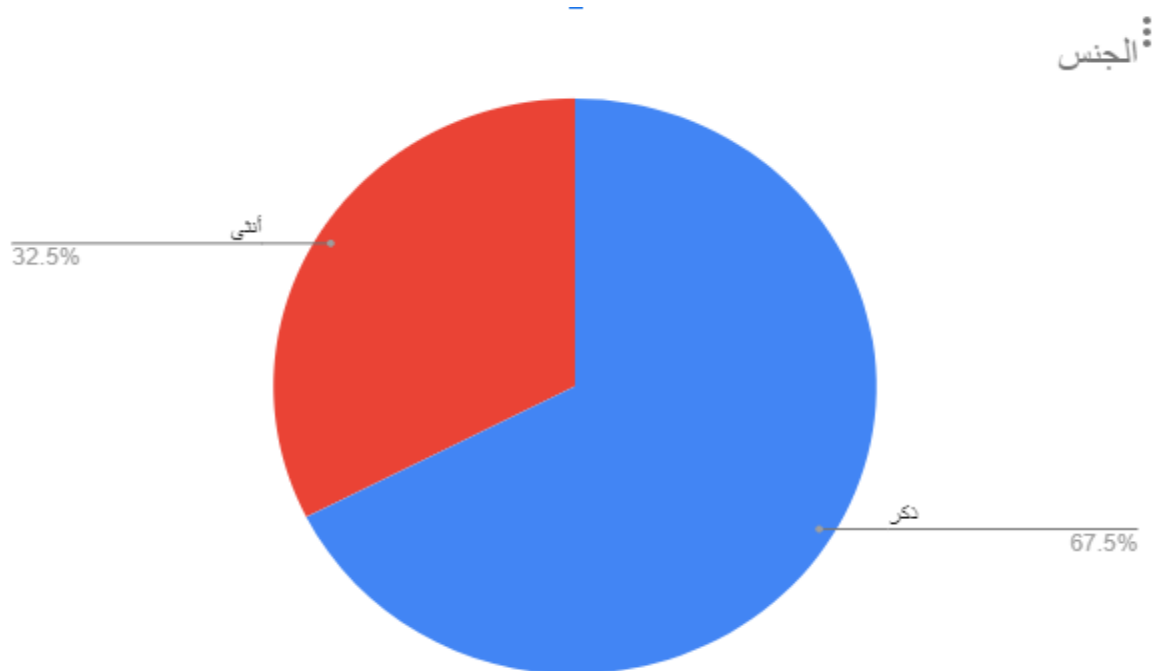


Figure 3.2.2

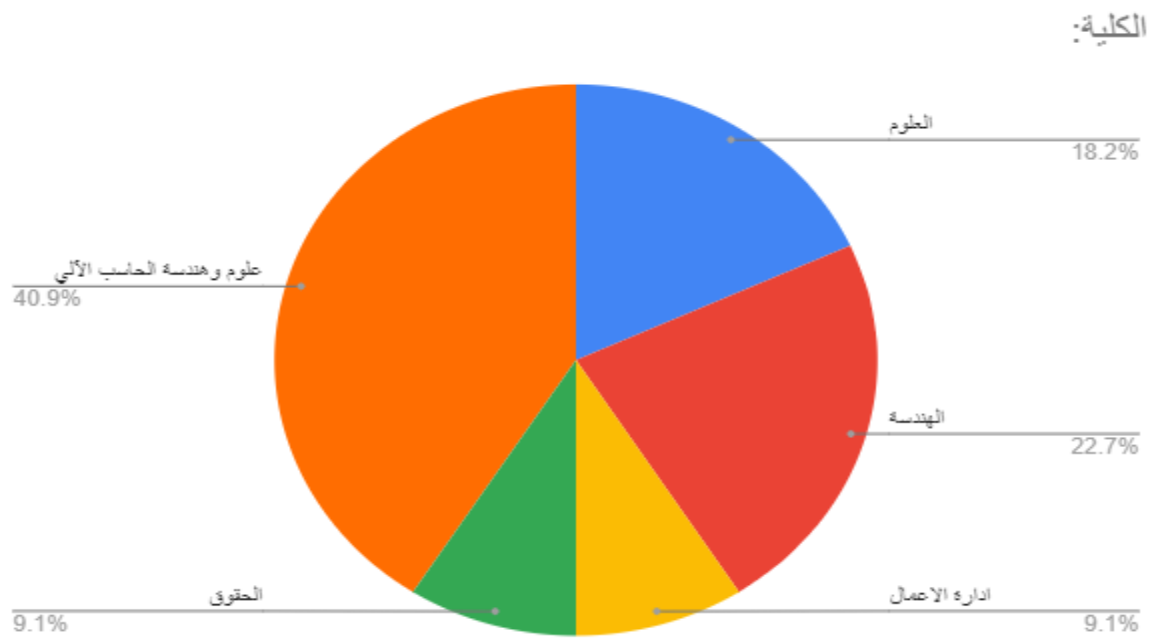


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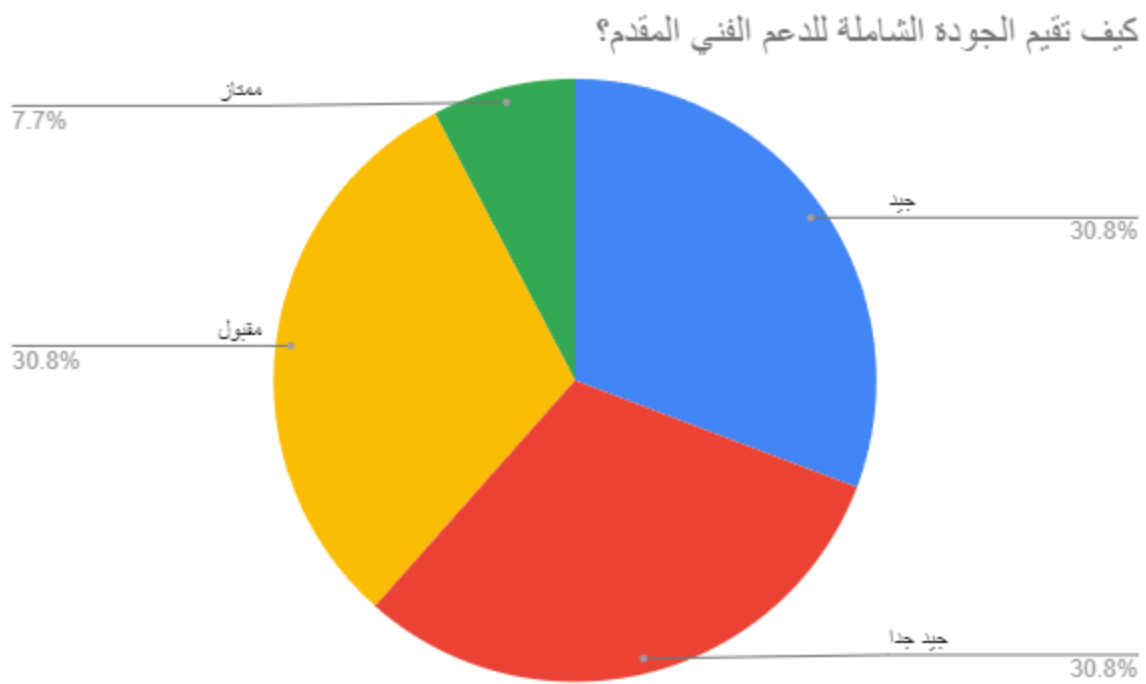


Figure 3.2.4

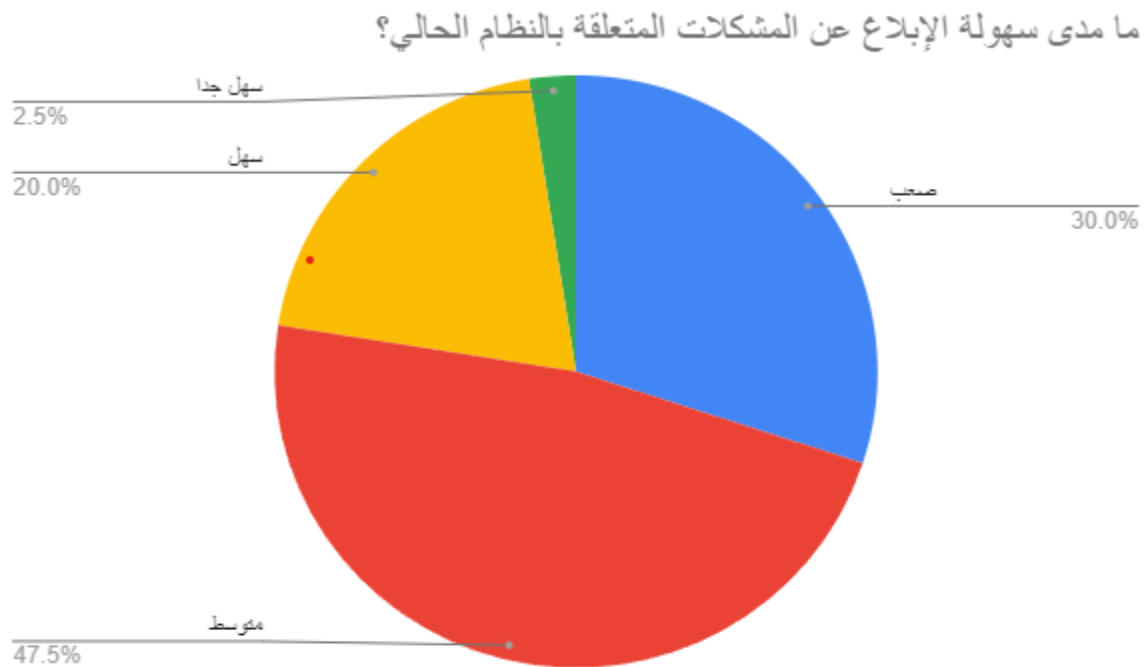


Figure 3.2.5

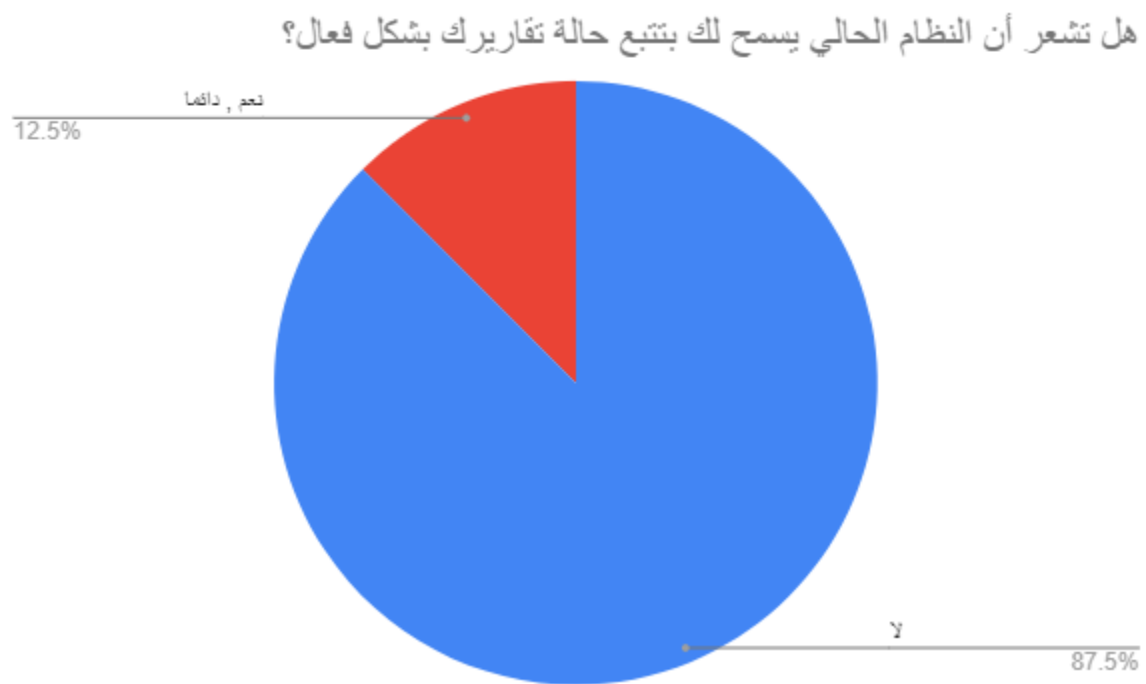


Figure 3.2.6

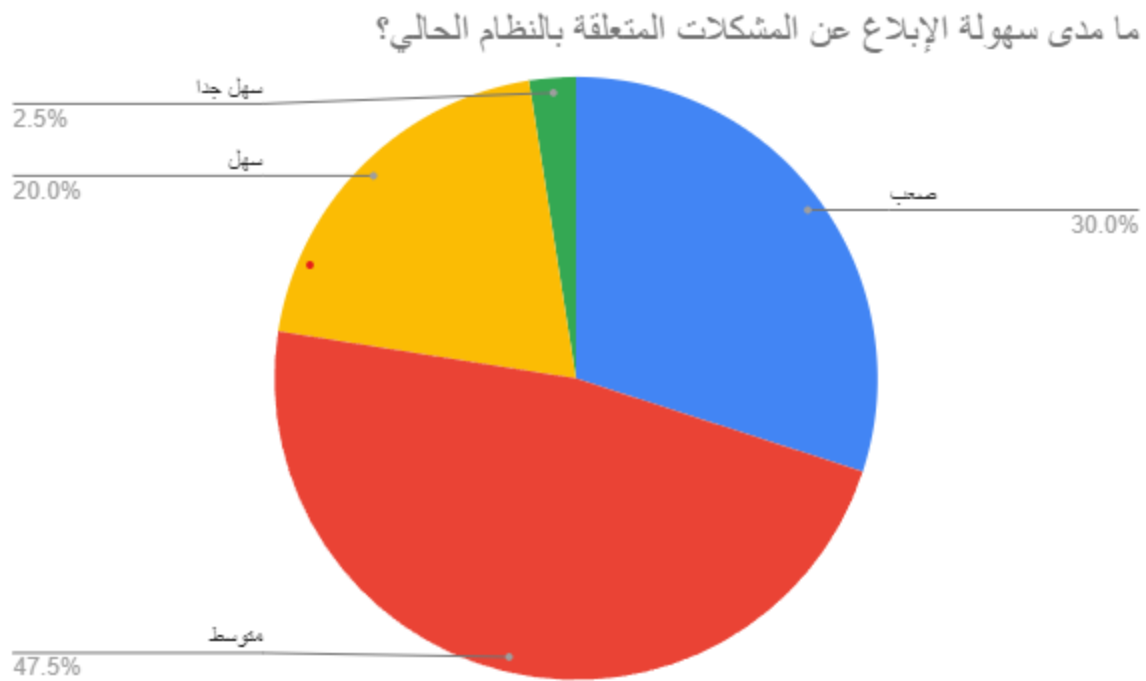


Figure 3.2.7

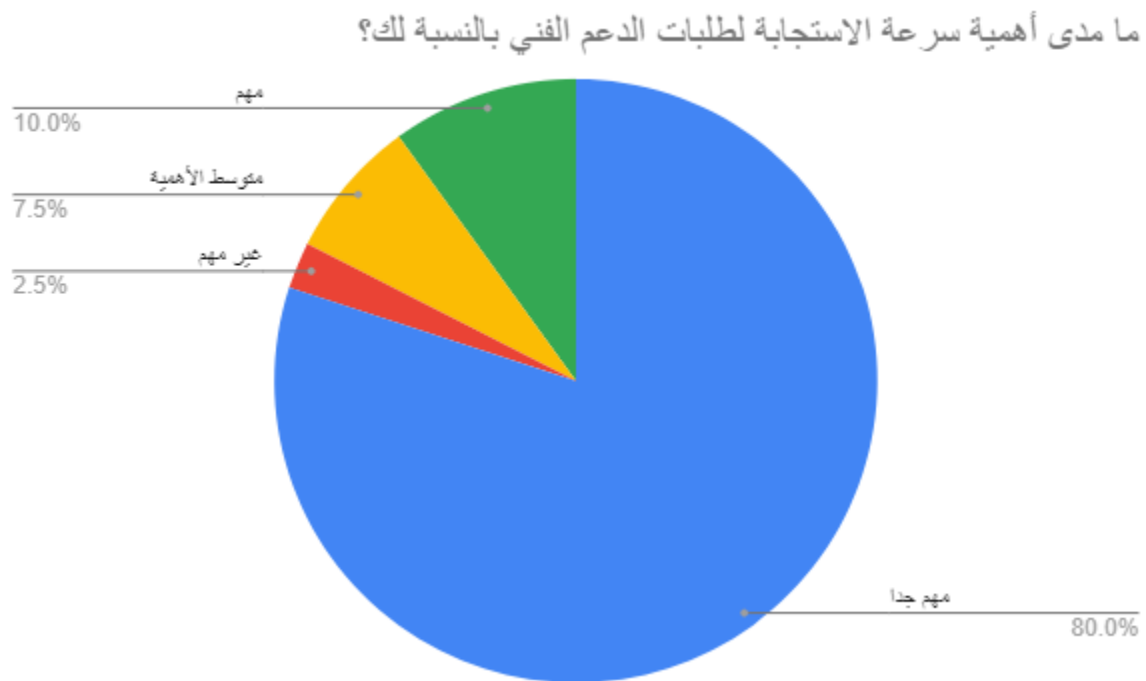


Figure 3.2.8



Figure 3.2.9

هل لديك أي اقتراحات لتحسين نظام الدعم الفني؟

11 responses

تخصيص فريق للدعم الفني متوفر على مدار الساعة وبناء نظام فعال يساهم في متابعة التقارير ومعرفة حالتها وسرعة إنقاذها

نعم تطوير البلاك بورد والانترنت في الجامعة والنظام الأكاديمي ووضع الدرجات للاختبارات التورية والنهائية في النظام الاكاديمي

يهمني حل المشكلة اكثر من سرعة الرد

اضافه شكاوى المشاكل التقنية، مثل الايميل

سرعه الرد وتحسين التواصل مع الطلاب

يرجى التجاوب مع البلاغات المقدمة

لا يوجد

سرعة الاستجابة و تسهيل واجهة المستخدم

نظام الدعم الفني يركز على خدمة العميل وسرعة الاستجابة لمشاكله او استفساراته التي تتطلب المساعدة

Figure 3.2.9.1 Suggestions

User Management System Guide

3.3.2 Functional Requirements

Requirement	Description	Roles Affected
User Registration	Users should be able to create an account by providing necessary details such as name, email, and password.	Students, Faculty
User Login	Users must be able to log into the system using their registered email and password.	Students, Faculty, Technicians
Role Assignment	Supervisors can assign or modify roles for users based on their responsibilities.	Supervisors
View User Activity	Supervisors should be able to view logs of user activities within the system for auditing purposes.	Supervisors
Notifications	Users should receive notifications for important system updates, role changes, or new features.	Students, Faculty, Technicians
Data Export	Supervisors can export user data reports in various formats for analysis and compliance purposes.	Supervisors

Table 3.3.2 functional requirement

3.3.2 Functional Requirement Details

User Registration

New users sign up by providing name, email, and password. The system validates email uniqueness/format and password strength. A confirmation email is sent for verification.

User Login

Secure access via email/password. Includes failed-attempt logout. Successful login redirects users to role-specific dashboards.

Role Assignment

Supervisors assign/modify user roles via an interface, ensuring compliance with policies.

View User Activity

Supervisors monitor user actions with filters (date, action, user) for auditing and compliance.

Access Control

Role-based restrictions ensure users only access permitted features, protecting sensitive data.

Notifications

Users receive alerts for role changes, system updates, or new features.

Data Export

Supervisors export user reports (CSV/Excel) for analysis and regulatory compliance.

3.3.3 Non-Functional Requirements

Requirement	Purpose	Importance
Performance	The system should respond to user actions within a few seconds to ensure a smooth user experience.	High - Affects user satisfaction and system usability.
Scalability	The system must accommodate too many users without performance degradation.	High - Ensures the system can grow with user demand.
Security	The system should implement encryption for sensitive data, including passwords and personal information.	Critical - Protects user data from unauthorized access and data breaches.
Reliability	The system should ensure high uptime, allowing users to access services consistently.	High - Maintains user trust and satisfaction with availability.
Usability	The interface should be intuitive and accessible, allowing users to navigate easily.	Medium - Enhances user experience and reduces training time.
Maintainability	The system should be modular, allowing for easier updates and maintenance without downtime.	Medium - Facilitates cost-effective long-term support and updates.
Availability	This platform must operate available online 24/7, so any user can access it from anywhere, at any time.	High - Maintains user trust and satisfaction for the service
Extensibility	The platform must be designed and developed to allow future expansion and improvements.	Medium it will serve a great purpose in the future

Table 3.3.3 non-functional requirement

3.3.4 Non-Functional Requirement Details

Performance

Performance refers to how quickly the system reacts to user inputs. For example, actions like logging in or submitting a form should be completed within two seconds. Fast response times are essential to keep users satisfied and engaged, while slow performance may result in frustration and abandonment of the system.

Scalability

Scalability is the system's capacity to manage increased usage without slowing down or crashing. It should be able to support up to 10,000 users at the same time, especially during high-demand periods like course registrations. This ensures the system remains reliable as the number of users grows.

Security

Security is critical, especially when the system deals with private user data. Data such as passwords and personal information must be encrypted to prevent unauthorized access. A secure system builds trust with users and helps avoid risks like data breaches, legal issues, and reputational damage.

Reliability

Reliability means the system works as expected without frequent failures. A target uptime of 99.9% ensures users can access the system whenever they need to. Consistent availability is crucial for keeping users confident in the system's dependability.

Usability

Usability focuses on how easy the system is to use. A user-friendly interface helps people complete tasks quickly and with minimal training. When users can navigate the system smoothly, it improves their overall experience and encourages repeated use.

Maintainability

Maintainability involves designing the system in a way that makes it simple to update and fix over time. Using a modular structure helps developers apply changes without disrupting the whole system. This makes it easier to keep the system running smoothly and up to date.

Extensibility

Extensibility ensures that the system can be upgraded or expanded in the future. By using modular components and APIs, new features can be added without reworking existing parts. Collecting user feedback can also help guide future improvements.

Availability

Availability ensures that the system is accessible at all times, from any location. This is achieved by having backup systems, distributing user load across servers, and continuously monitoring performance. Consistent availability builds trust and reliability among users.

3.4 Requirements Specification

In this section, this project is using a use-case diagram for the proposed project to clarify the functions of the system and descriptions for each of the functions.

3.4.1 Use case diagram

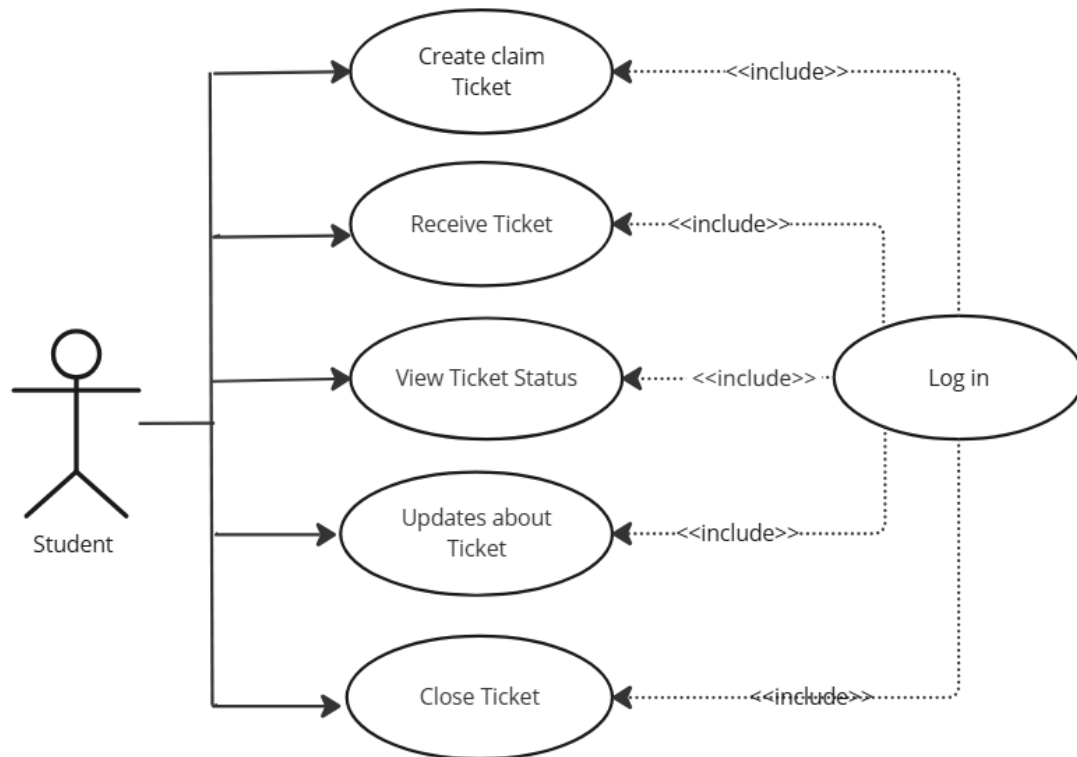


Figure 3.4.1 student use-case diagram

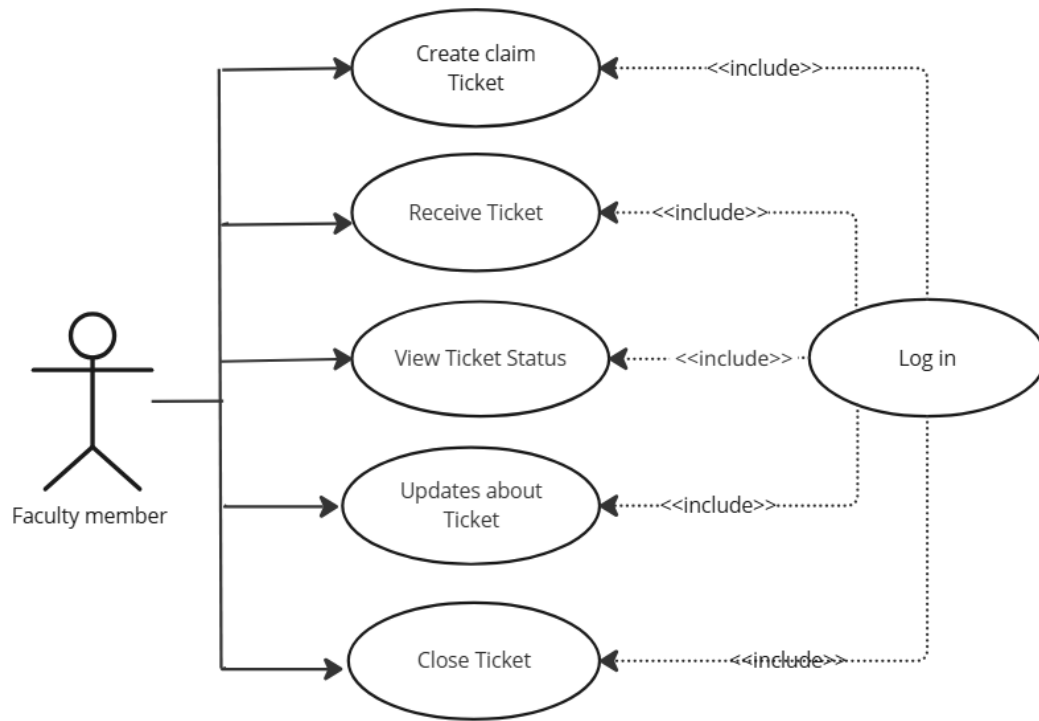


Figure 3.4.2 Faculty member use-case diagram

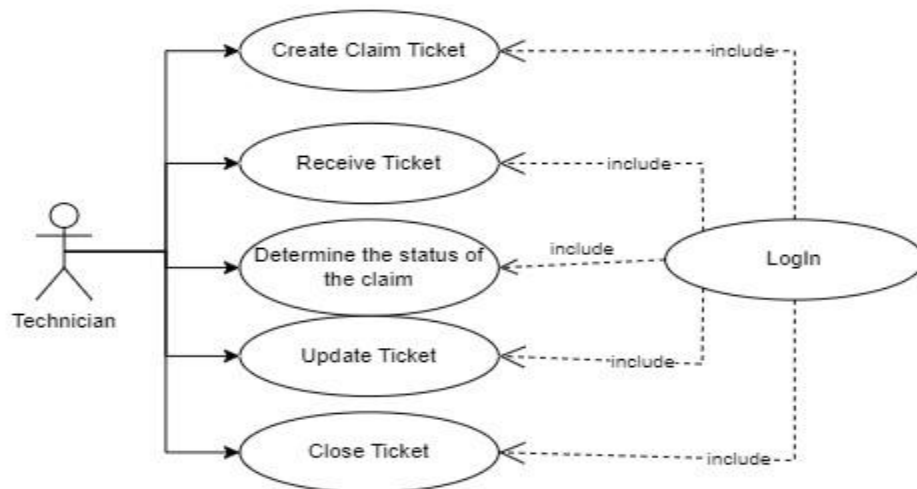


Figure 3.4.3 Technician use-case diagram

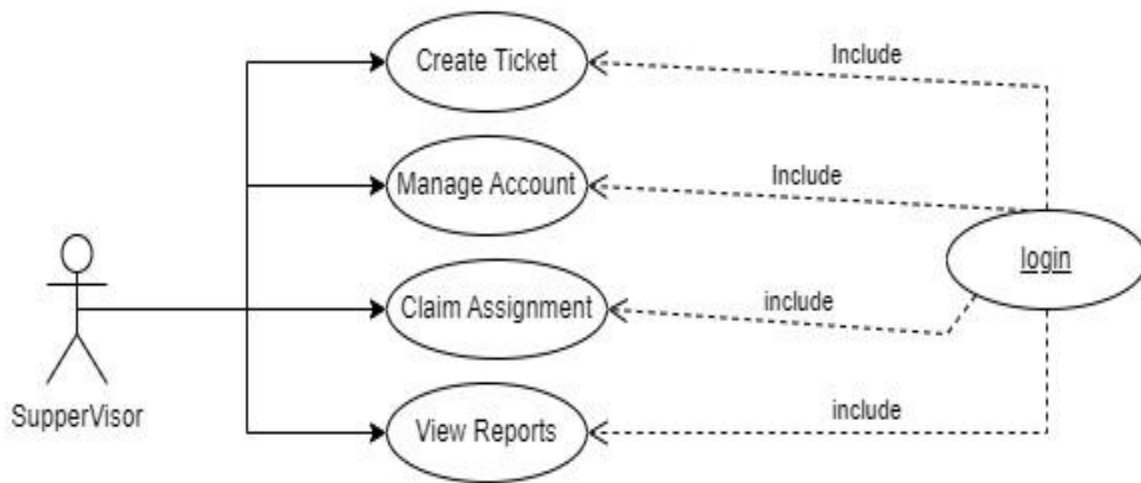


Figure 3.4.4 Supervisor/Admin use-case diagram

3.4.2 Use-case descriptions

Use Case: Report Technical Issue

The following table outlines the use case for the 'Report Technical Issue' feature within the Technical Support System. This feature allows both students and faculty members to report any technical issues they encounter while using the system.

Actors	Pre-condition	Flow of Events	Post-condition
Student, Faculty Member	User must be registered and logged in to the technical support system.	1. User navigates to the technical support website. 2. User clicks on "Report Issue." 3. A reporting form is displayed. 4. User fills out the form with details of the issue. 5. User reviews the information. 6. User clicks "Submit." 7. System validates entered data. 8. System stores the report. 9. System sends a confirmation message to the user.	Success: Issue successfully stored with confirmation. Failure: Prompts to correct incomplete or invalid information.

Actors	Pre-condition	Flow of Events	Post-condition
		10. System updates the issue tracking database.	

Description of the Use Case

- Allows students and faculty members to report technical issues they encounter while using the system.
- Users must be registered and logged in to the technical support system to report an issue.

Use Case: Track Issue Status

The following table outlines the use case for the 'Track Issue Status' feature within the Technical Support System. This feature allows users to monitor the status of the technical issues they have reported.

Actors	Pre-condition	Flow of Events	Post-condition
Student, Faculty Member	User has reported at least one issue and is logged in.	1 User accesses the technical support system. User selects "Track Issues." The system displays the status of reported issues. .	User can view the current status of their reported issues..

Description of the Use Case

- Allows users to check the current status of the technical issues they have reported.
- Users must have reported at least one issue and be logged in to the system.

Use Case: Assign Issue to Technician

The following table outlines the use case for the 'Assign Issue to Technician' feature within the Technical Support System. This feature enables supervisors to effectively assign reported issues to the appropriate technicians for resolution.

Actors	Pre-condition	Flow of Events	Post-condition
Supervisor	Supervisor must be logged into the system.	1- Supervisor accesses the list of reported issues. 2- Supervisor selects an issue to assign. 3- Supervisor chooses a technician based on availability and expertise. 4- Supervisor assigns the issue to the selected technician. 5- System updates the issue status to "Assigned."	The issue is assigned to a technician, and the technician is notified.

Description of the Use Case

- Enables supervisors to assign reported technical issues to the appropriate technicians for resolution.
- Supervisors must be logged in to the administrator interface and have unassigned issues available.
- The system updates the issue status and notifies the assigned technician.

Use Case: Update Issue Status

The following table outlines the use case for the 'Update Issue Status' feature within the Technical Support System. This feature allows technicians to update the status of reported issues and provide resolution details as necessary.

Actors	Pre-condition	Flow of Events	Post-condition
Technician	Technician is logged in and has been assigned to an issue..	1-Technician accesses their assigned issues. 2-Technician selects an issue to update. 3-Technician updates the status (e.g., In Progress, Resolved). 4-Technician provides additional notes if necessary. 5-System saves the updated status and notes.	Issue status is updated, and users are notified of the change.

Description of the Use Case

- Allows technicians to update the status of the issues assigned to them and provide resolution details.
- Technicians must be logged in to the technician interface and have assigned issues in their queue.
- The system updates the issue status and notifies the user who reported the issue.

Use Case: Generate Reports

The following table outlines the use case for the 'Generate Reports and Analytics' feature within the Technical Support System. This feature enables supervisors and IT administrators to generate comprehensive reports and analyze data, facilitating informed decision-making.

Actors	Pre-condition	Flow of Events	Post-condition
Supervisors	Supervisor is logged in..	1-Supervisor navigates to the reporting section of the system. 2-Supervisor selects the type of report (e.g., issues resolved, average response time). 3- Supervisor specifies the date range and parameters for the report. 4-System generates the report. 5-Supervisor can download or print the report.	Report is generated and made available for download or printing.

Description of the Use Case

- Enables supervisors and IT administrators to generate reports related to the technical support system.
- Users must be logged in to the administrator interface and have sufficient data available in the system.

3.5 Development Methodology

The Agile methodology has been selected for executing the process, as it allows for continuous feedback throughout all phases of the project. It is also an iterative process, making it collaborative and repeatable.

Agile consists of six main phases as follows:

1. **Project Scope Definition:** The project and its scope should be defined early on and communicated to all team members.
2. **Planning Phase:** This phase involves outlining the project plans, including resources, time budget, and potential risks.
3. **Requirements Analysis:** Requirements must be established based on the project's objectives and goals.
4. **Design Phase:** This step includes both the logical and physical design of the proposed model, based on the requirements gathered in the analysis phase.
5. **Implementation Phase:** This phase relies on the design specifications, with each team member responsible for a different component of the implementation.
6. **Testing Phase:** In this phase, the work of each group is combined and tested to eliminate any possible faults, errors, or redundancies.

Agile is one of the most popular approaches to project management because it is flexible, it is adaptable to changes, and it encourages customer feedback.

Many teams embrace this approach to achieve the following benefits of Agile:

- **Rapid progress:** By effectively reducing the time it takes to complete various stages of a project, teams can elicit feedback in real time and produce working prototypes or demos throughout the process.
- **Continuous improvement:** As an iterative approach, the Agile project management methodology allows teams to chip away at tasks until they reach the best end result.



Figure 3.5.1.1 Development Methodology

3.6 Summery

focuses on system analysis, clarifying objectives through diagrams and detailing requirements for the technical support system. It begins with user feedback collected via a questionnaire from 40 participants, highlighting acceptance insights for the new system.

Key functional requirements include user registration, login, password recovery, role assignment, activity management, notifications, and data export. Non-functional requirements cover performance, scalability, security, and reliability.

The Agile methodology is chosen for project execution, facilitating continuous feedback across six phases, from defining project scope to testing. Overall, the chapter provides a comprehensive overview of the requirements and processes essential for developing an effective technical support system for the educational institution.

Chapter 4: System Design

4.1 Introduction

System design is a crucial step in the software development process, as it defines the system's architecture, structure, and interaction models. It provides the framework for how users interact with the system and how the components collaborate. This chapter will outline the architectural design, static structural models such as class diagrams and ERDs, and dynamic models like sequence diagrams. These models collectively describe the logical, physical, and behavioral aspects of the system.

4.2 Architectural Design

The proposed technical support system is designed with a three-layer architecture to ensure scalability, flexibility, and maintainability. The layers include:

1. **Presentation Layer:** Provides a user-friendly interface accessible via web browsers and mobile devices. This layer interacts with users (students, faculty, technicians, and supervisors) and captures their input.
2. **Application Layer:** Contains the business logic, which processes user inputs, interacts with the database, and executes operations like issue reporting, status tracking, and technician assignment.
3. **Database Layer:** Manages all data storage, retrieval, and relationships using an organized schema. The system ensures secure and efficient handling of user and issue data.



Figure 4.2.1

4.3 Object Oriented Design

4.3.1 Structural Static Model

The ERD showcases the logical database design, including primary and foreign keys and the relationships between entities. Entities such as Users, Issues, Rooms, and Reservations are defined with their attributes and linked based on their interactions.

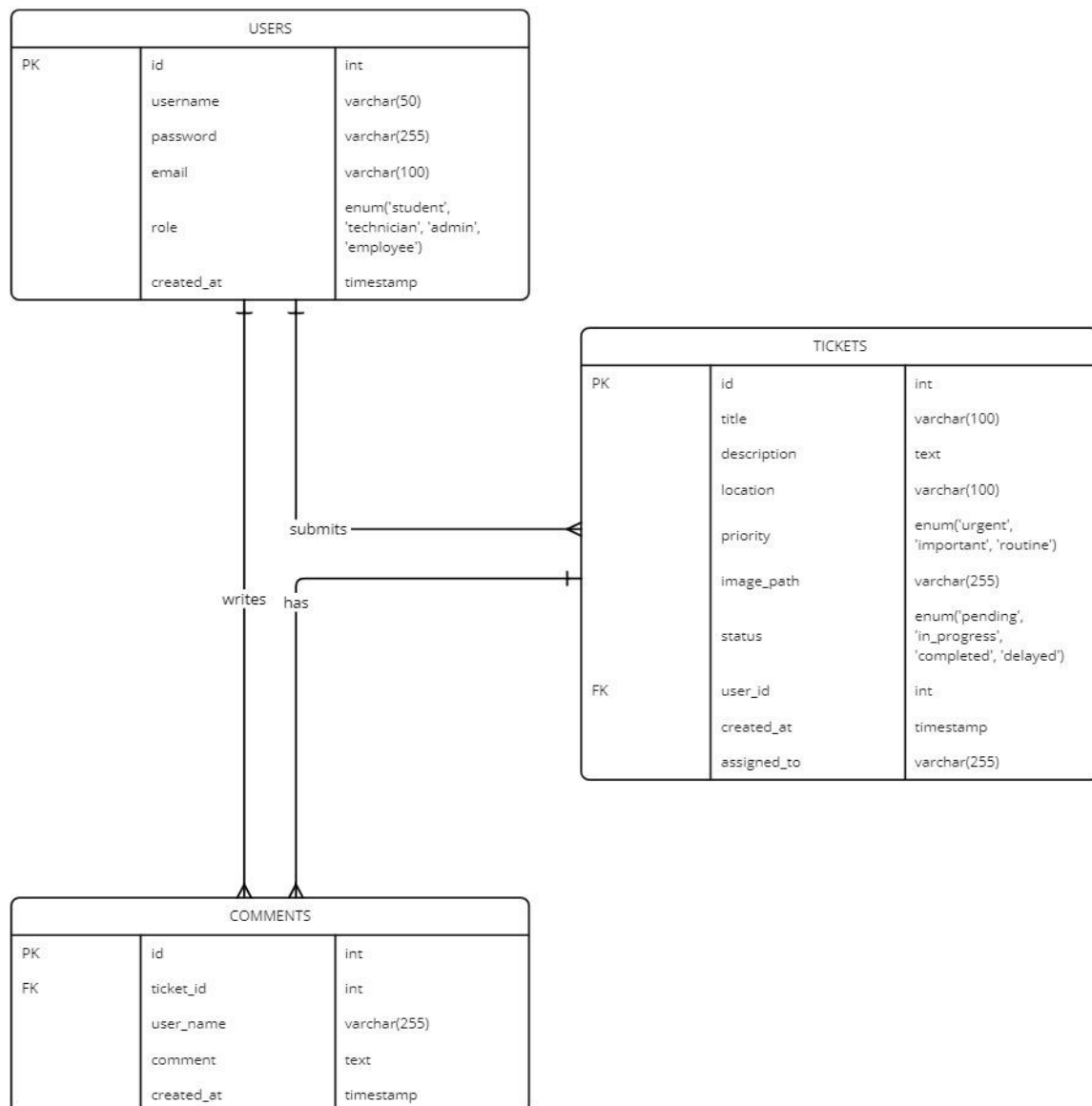


Figure 4.3.1ERD

4.3.2 Dynamic Model

4.3.2.1 Sequence Diagram

A sequence diagram is a type of interaction diagram that illustrates the order in which a group of objects interact with one another. It is also referred to as an event diagram or event scenario.

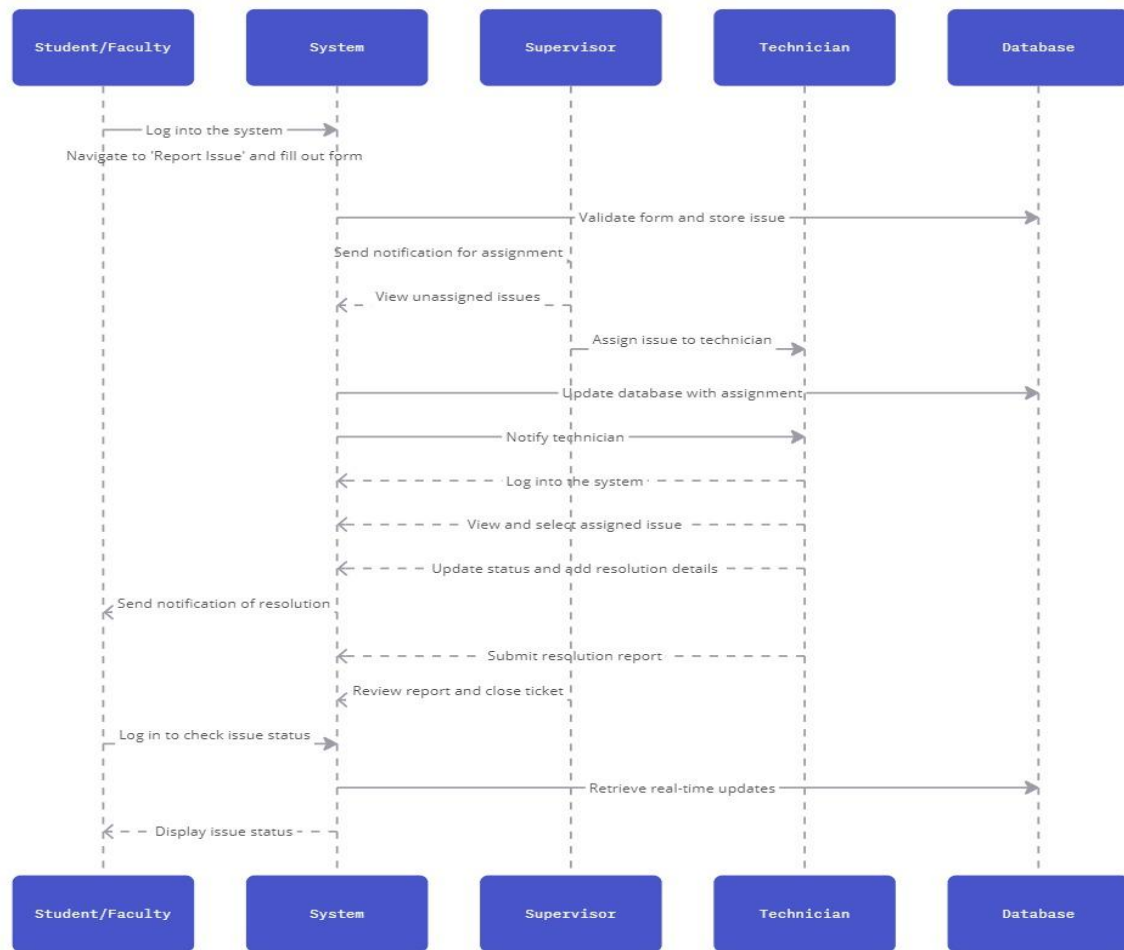


Figure 4.3.2 Sequence Diagram

4.4 Data Modeling

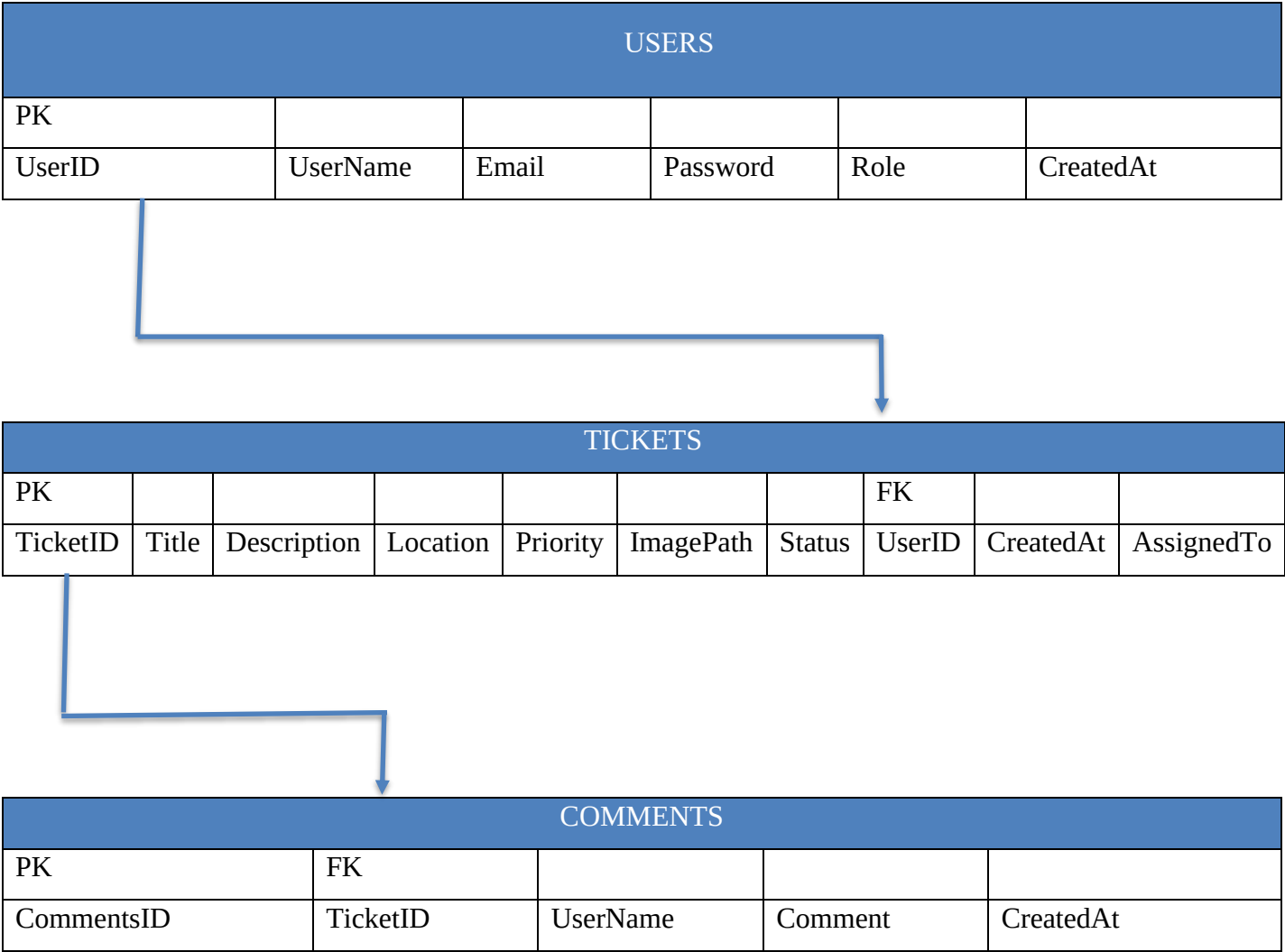
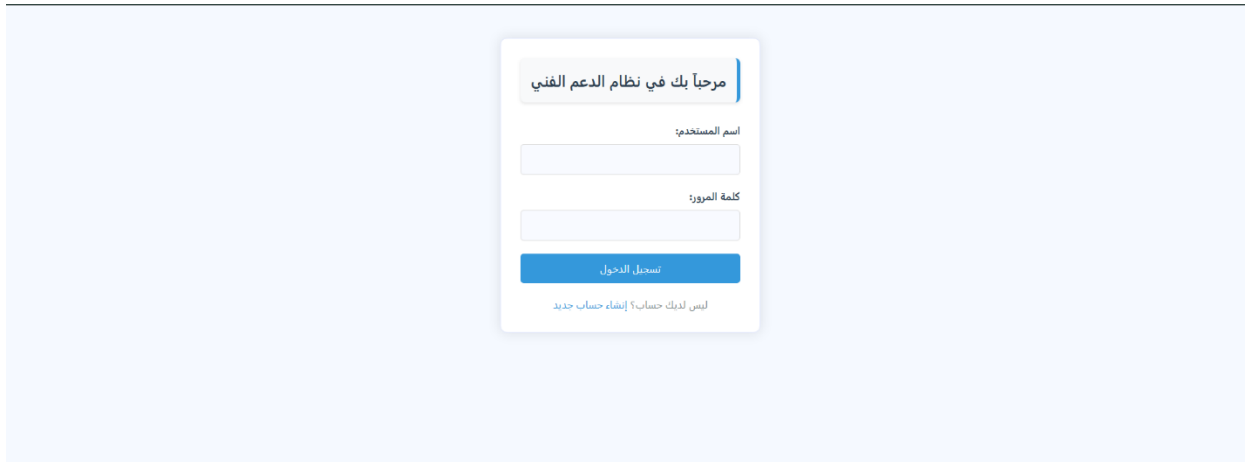


Figure 4.4 Data Modeling

4.5 User Interface Design



مرحباً بك في نظام الدعم الفني

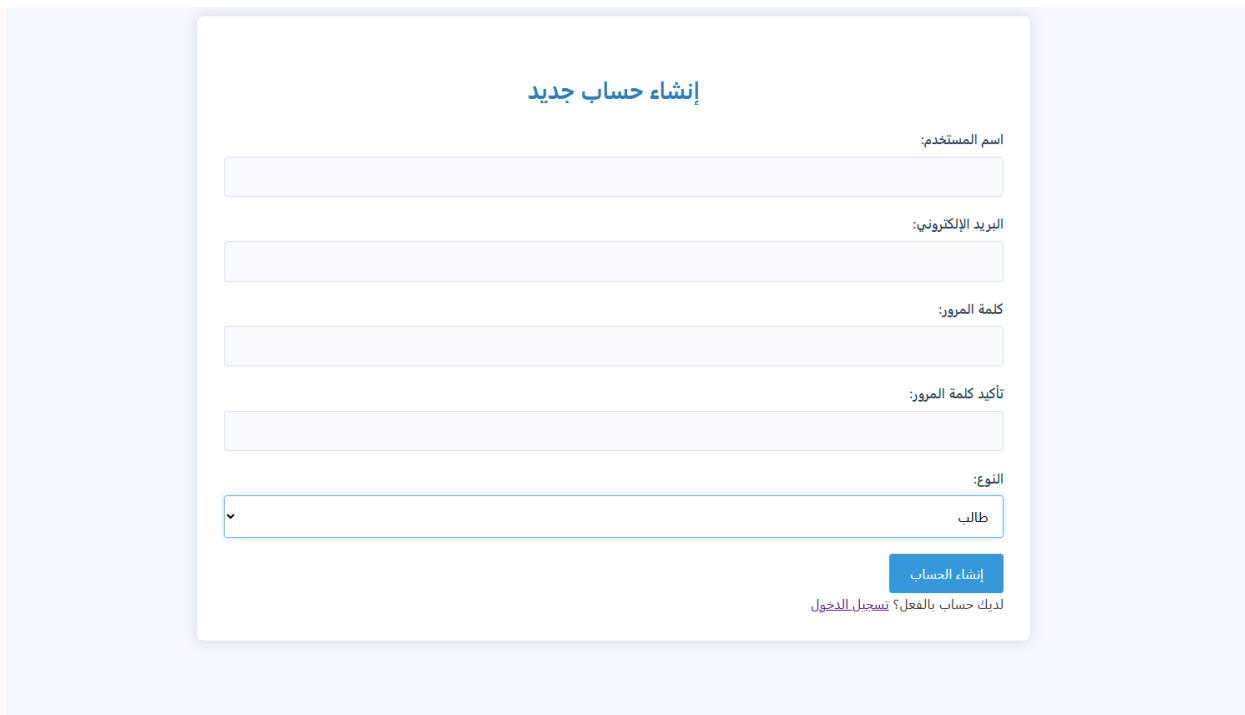
اسم المستخدم:

كلمة المرور:

تسجيل الدخول

ليس لديك حساب؟ إنشاء حساب جديد

Figure 4.5.1 log in page UI



إنشاء حساب جديد

اسم المستخدم:

البريد الإلكتروني:

كلمة المرور:

تأكيد كلمة المرور:

النوع:

طالب

إنشاء الحساب

لديك حساب بالفعل؟ تسجيل الدخول

Figure 4.5.2 Create a new user

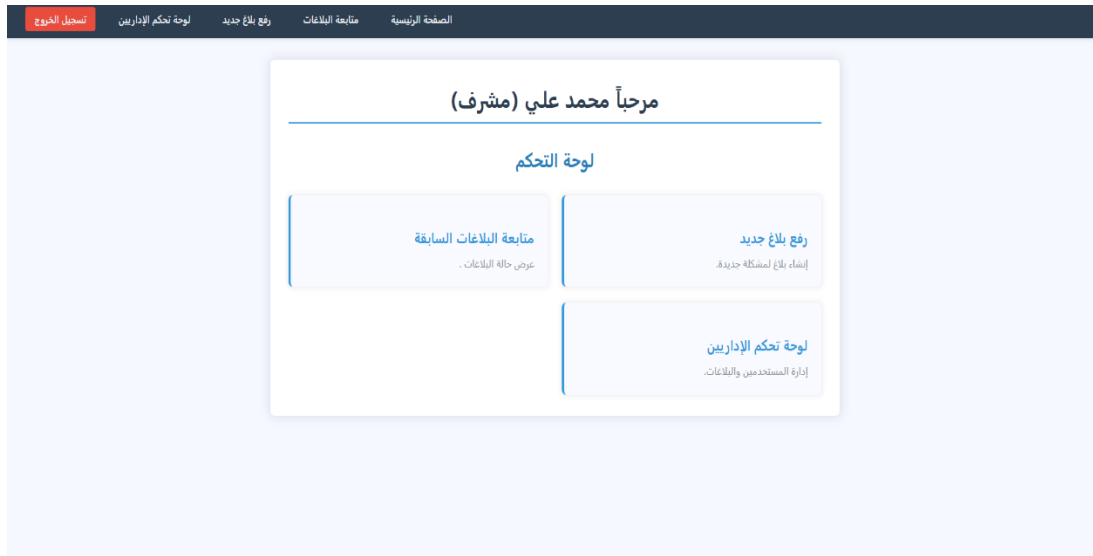


Figure 4.5.3 Admin interface



Figure 4.5.4 Technician interface

[تسجيل الخروج](#)[رفع بلاغ جديد](#)[متابعة البلاغات](#)[الصفحة الرئيسية](#)

رفع بلاغ جديد

عنوان البلاغ:

وصف المشكلة:

مكان الجهاز:

تصنيف البلاغ:

عاجل

إرفاق صورة (اختياري):

إرفاق ملف

رفع البلاغ

Figure 4.5.5 creating a new ticket

[تسجيل الخروج](#)[رفع بلاغ جديد](#)[متابعة البلاغات](#)[الصفحة الرئيسية](#)

متابعة البلاغات

[جميع البلاغات](#)[بلاغات الخاصة](#)

عنوان البلاغ	التصنيف	الحالة	الوقت	التفاصيل
AC issue	urgent	completed	19:53:45 2025-04-29	عرض التفاصيل
AC issue	important	in_progress	19:56:38 2025-04-29	عرض التفاصيل
meeting tables	important	completed	20:45:24 2025-04-29	عرض التفاصيل
Projector issue	urgent	completed	15:36:46 2025-05-05	عرض التفاصيل

تفاصيل البلاغ

Projector issue	عنوان البلاغ:
Lag and constant shut down	وصف البلاغ:
urgent	التصنيف:
completed	حالة البلاغ:
15:36:46 2025-05-05	تاريخ الإنشاء:
Technician2	الغني المسؤول:
	صورة البلاغ:

التعليقات:

Technician2
تم الحل
19:09:34 2025-05-05

أضف تعليقاً...

إضافة تعليق

Figure 4.5.6 Reports follow up for supervisor

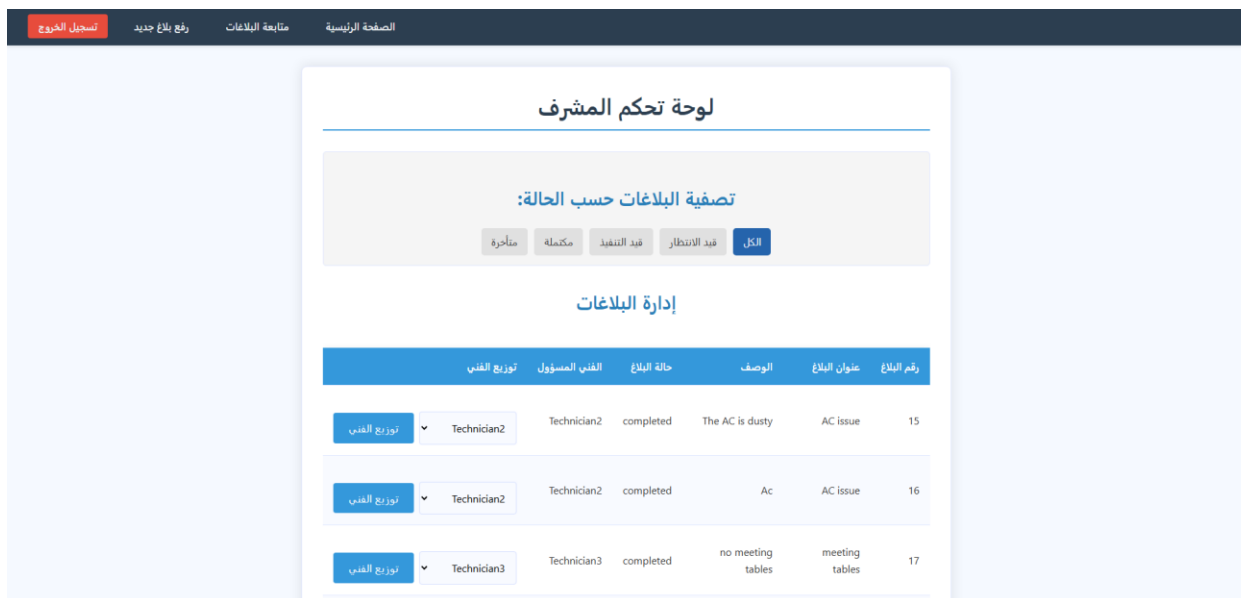


Figure 4.5.7 Reports management

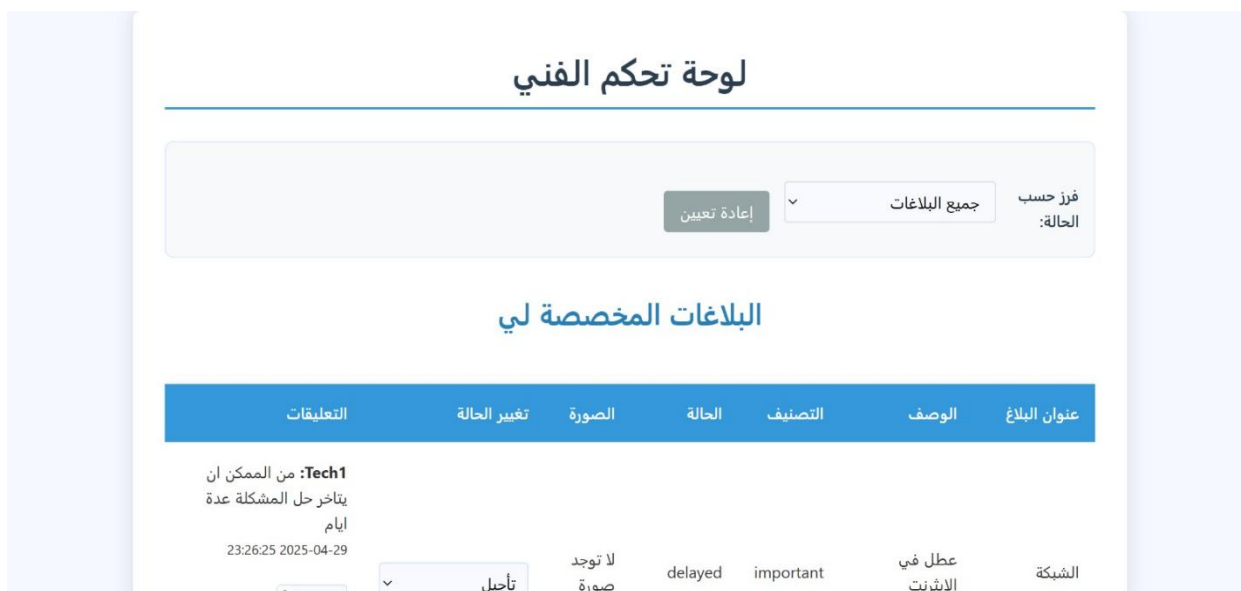


Figure 4.5.8 Technician reports management



Figure 4.5.9 Faculty member interface

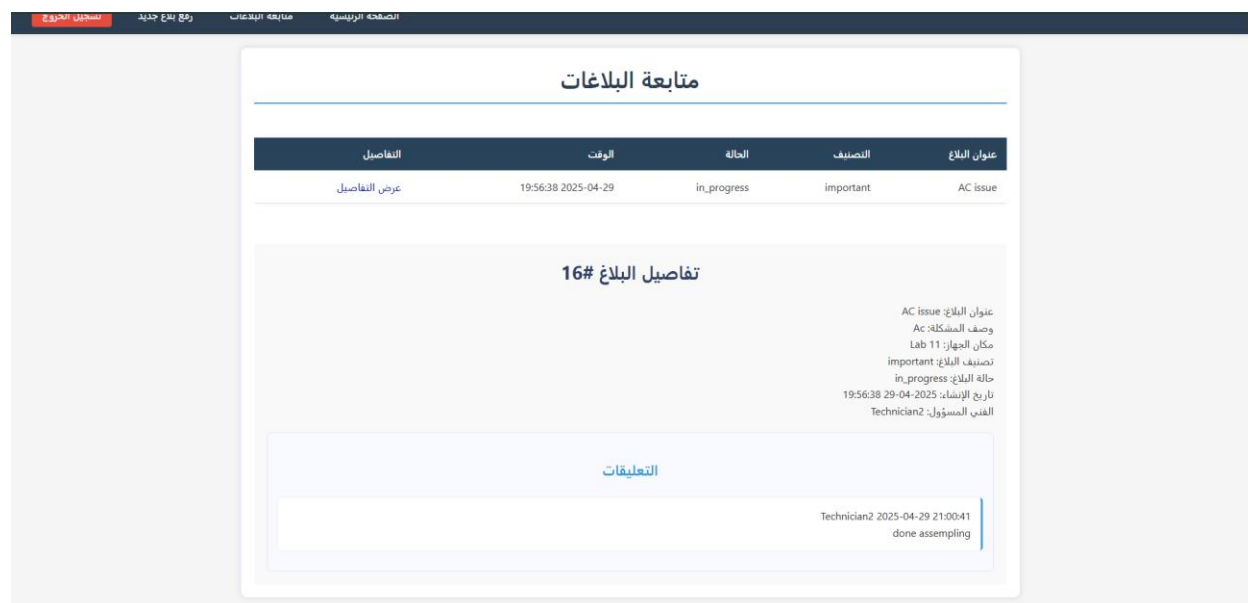


Figure 4.5.9.0 Follow up on posted tickets for Students/Faculty members

4.6 Summary

Chapter 4 focuses on the system design of the proposed technical support system for Taibah University. It outlines the architectural and object-oriented designs necessary for developing a robust and efficient system.

Key components covered in this chapter include:

Architectural Design: This section presents the overall structure of the system, detailing how different components interact with each other to facilitate the technical support processes.

Object-Oriented Design: The chapter elaborates on the use of object-oriented principles to create a modular system, enhancing maintainability and scalability. Diagrams such as class diagrams illustrate the relationships between various classes and objects within the system.

Data Modeling: This portion emphasizes the importance of data organization, including how data will be stored, accessed, and managed.

User Interface Design: The design of user interfaces is addressed, focusing on creating an intuitive and user-friendly experience for all stakeholders, including students, faculty, and technical staff.

Summary of Key Features: The chapter concludes with a summary of critical features and functionalities that will be implemented to ensure the system meets its objectives effectively.

Overall, Chapter 4 lays the groundwork for the development phase by providing comprehensive design specifications that align with the project's goals of improving technical support at Taibah University.

Chapter 5: System Implementation

5.1 Introduction

System implementation is a critical phase in the software development life cycle, where theoretical designs and plans are transformed into operational systems. This phase involves deploying software applications, configuring hardware, and ensuring that all components work seamlessly together. Effective implementation requires meticulous planning, resource allocation.

During this process, teams conduct rigorous testing to identify and resolve any issues before the system goes live.

5.2 Tools and Languages

	We have chosen to use Visual Studio Code (VS Code) as code editor for the project due to its powerful capabilities and multiple features. This open-source editor supports various programming languages, making it ideal for meeting the diverse needs of the project. VS Code is equipped with features such as syntax highlighting, debugging, and version control integration, which facilitate the development process. Additionally, it includes an integrated terminal and an extensions marketplace, allowing me to customize my work environment according to the project requirements.
	XAMPP is chosen for projects using PHP because it provides an integrated and easy-to-install development environment. It includes Apache server and MySQL database, simplifying the setup process. With built-in support for PHP, it allows for local testing of code before deployment, helping to identify and fix issues. Additionally, XAMPP's user-friendly interface enhances the overall development experience, boosting efficiency in project development.
	Choosing PHP as a back-end language offers several advantages. PHP is widely used for web development due to its simplicity, flexibility, and strong community support. It excels in creating dynamic web pages and integrates seamlessly with databases, making it ideal for data-driven applications. PHP is fully compatible with XAMPP, which provides a ready-to-use environment that includes the Apache server and MySQL database. This compatibility allows developers to easily run and test PHP applications locally, streamlining the

	development process. With XAMPP, developers can leverage PHP's features effectively, facilitating rapid application development and deployment.
	SQL (Structured Query Language) is a programming language used for managing databases and offers a set of tools that facilitate data handling. In a project, SQL plays a vital role by enabling efficient data management, allowing for the creation of new databases, defining tables, and storing information in an organized manner. It also makes it easy to retrieve the required data, supporting effective information display for users.
	HTML (Hypertext Markup Language) is used to create the structure of the page and organize content. It defines essential elements such as headings, paragraphs, links, and images, helping to arrange information in a logical and understandable manner. Its clear structure enhances the user experience and facilitates navigation. On the other hand, CSS (Cascading Style Sheets) adds aesthetics to these interfaces. It allows for the formatting of content, such as defining colors, fonts, spacing, and layouts. With CSS, it's possible to achieve an attractive and cohesive design, increasing the project's appeal and effectiveness. By using HTML and CSS together, projects can deliver integrated and user-friendly interfaces, improving user interaction and contributing to overall success.

Table 5.2 tools and languages

5.3 Mapping Design to Implementation

Architectural design: The successful implementation of the proposed three-layer architecture in the project ensures a robust technical support system. The presentation layer offers an intuitive interface for users, enhancing accessibility across devices. The application layer effectively processes inputs and manages business logic, facilitating smooth operations like issue reporting and tracking. Finally, the database layer securely handles data storage and retrieval, ensuring efficient management of user and issue information.

Sequence diagram: The system explains the process in a simplified manner, detailing how requests are processed and tracked. The user starts by submitting a request through the system, which is then directed to the supervisor for review and assignment to the appropriate technician. The status of the request is updated in the system, allowing the user to monitor progress. However, the notification system has not yet been implemented due to the lack of email linkage for the user.

ERD diagram: The ERD model illustrates the structure of a database consisting of three main entities: users, tickets, and comments.

- Users contains information such as username, password, email, and role (e.g., student, employee, admin).
- Tickets includes details like title, location, priority, and ticket status, along with the associated image path.
- Comments records the users' notes related to tickets, referencing the associated ticket and the username of the commenter.

The model clarifies the relationships between these entities, showing how users create tickets and comment on them, reflecting the interaction of different elements within the system.

Data Modeling: The Data Modeling model is the process of designing the data structure in a specific system and relies heavily on the ERD model. Data Modeling begins by identifying the main entities. After identifying the entities, the model clarifies the attributes associated with each entity. Thus, both models work together to provide a comprehensive view of how data is organized and interacts, with Data Modeling assisting in planning and design, while the ERD visually illustrates the relationships.

User interface design: The user interfaces in the project were developed using HTML and CSS. Different interfaces were designed for each user based on their roles, such as admin, student, technician, and employee. This customization ensures a user experience that meets the needs of each category.

5.4 Most important codes

```

session_start();
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    include 'config.php'; // الاتصال بقاعدة البيانات
    $username = $_POST['username'];
    $password = $_POST['password'];

    $query = "SELECT * FROM users WHERE username = ?";
    $stmt = $conn->prepare(query: $query);
    $stmt->bind_param(types: "s", var: &$username);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows > 0) {
        $user = $result->fetch_assoc();
        if (password_verify(password: $password, hash: $user['password'])) {
            $_SESSION['user_id'] = $user['id'];
            $_SESSION['username'] = $user['username'];
            $_SESSION['role'] = $user['role'];
            header(header: "Location: dashboard.php");
        } else {
            $error = "خطأ في كلمة المرور.";
        }
    } else {
        $error = "اسم المستخدم غير موجود.";
    }
}

?>
<!DOCTYPE html>
<html lang="ar">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>تسجيل الدخول</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="login-container">
        <form method="POST">
            <label for="username">الاسم المستخدم</label>
            <input type="text" id="username" name="username" required>

            <label for="password">كلمة المرور</label>
            <input type="password" id="password" name="password" required>

            <?php if (isset($error)) { echo "<div class='error'>$error</div>"; } ?>

            <button type="submit">تسجيل الدخول</button>
        </form>
    </div>
</body>
</html>

```

Figure 5.4.1 Register code

```

<?php
include 'config.php'; // الاتصال بقاعدة البيانات
// التحقق من إرسال النموذج
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $username = $_POST['username'];
    $password = $_POST['password'];
    $confirm_password = $_POST['confirm_password'];
    $email = $_POST['email'];
    $role = $_POST['role'];

    // (global variable) $confirm_password
    if ($password != $confirm_password) {
        $error = "كلمة المرور لا تتطابق.";
    } else {
        // التحقق من كلمة المرور
        $hashed_password = password_hash(password: $password, algo: PASSWORD_DEFAULT);

        // التحقق من وجود المستخدم
        $query = "SELECT * FROM users WHERE username = ? OR email = ?";
        $stmt = $conn->prepare(query: $query);
        $stmt->bind_param(types: "ss", var: &$username, vars: &$email);
        $stmt->execute();
        $result = $stmt->get_result();

        if ($result->num_rows > 0) {
            $error = "البريد الإلكتروني أو اسم المستخدم موجود بالفعل.";
        } else {
            // إدخال البيانات
            $query = "INSERT INTO users (username, password, email, role) VALUES (?, ?, ?, ?)";
            $stmt = $conn->prepare(query: $query);
            $stmt->bind_param(types: "ssss", var: &$username, vars: &$hashed_password, $email, $role);

            if ($stmt->execute()) {
                header(header: "Location: index.php"); // التوجيه إلى الصفحة الرئيسية
                exit();
            } else {
                $error = "حدث خطأ أثناء التسجيل. الرجاء المحاولة مرة أخرى.";
            }
        }
    }
}

```

Figur5.4.2 log in code

```
//
// تسجيل الدخول
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $title = $_POST['title'];
    $description = $_POST['description'];
    $location = $_POST['location'];
    $priority = $_POST['priority'];
    $user_id = $_SESSION['user_id']; // رقم المستخدم الذي أنشأ التذكرة

    // التحقق من وجود المستخدم في قاعدة البيانات
    $check_user = $conn->prepare(query: "SELECT id FROM users WHERE id = ?");
    $check_user->bind_param(types: "i", var: &$user_id);
    $check_user->execute();
    $result_check = $check_user->get_result();

    // إذا لم يتم العثور على المستخدم في قاعدة البيانات، فاعرض رسالة خطأ
    if ($result_check->num_rows == 0) {
        echo "خطأ: المستخدم غير موجود في قاعدة البيانات";
        exit();
    }

    // التحقق من نوع الصورة
    $image_path = null; // متغير لتخزين مسار الصورة
    if (isset($_FILES['image']) && $_FILES['image']['error'] === UPLOAD_ERR_OK) {
        $image_name = $_FILES['image']['name'];
        $image_tmp_name = $_FILES['image']['tmp_name'];
        $image_extension = strtolower(string: pathinfo(path: $image_name, flags: PATHINFO_EXTENSION));
        $allowed_extensions = ['jpg', 'jpeg', 'png', 'gif'];

        if (in_array(needle: $image_extension, haystack: $allowed_extensions)) {
            $new_image_name = uniqid(prefix: 'img_', more_entropy: true) . "." . $image_extension;
            $upload_directory = 'uploads/';
            $image_path = $upload_directory . $new_image_name;

            // التحقق من وجود المجلد
            if (!file_exists(filename: $upload_directory)) {
                mkdir(directory: $upload_directory, permissions: 0777, recursive: true);
            }

            move_uploaded_file(from: $image_tmp_name, to: $image_path);
        } else {
            echo "خطأ: ملف الصورة غير صحيح";
            exit();
        }
    }
}
```

Figure 5.4.3 Create ticket code

```
//
// تسجيل الدخول
if ($role == 'admin') {
    if (isset($_GET['type']) && $_GET['type'] == 'my_tickets') {
        // عرض التذاكر الخاصة بالمستخدم
        $query = "SELECT * FROM tickets WHERE created_by = ?";
        $stmt = $conn->prepare(query: $query);
        $stmt->bind_param(types: "i", var: &$user_id);
    } else {
        // عرض جميع التذاكر
        $query = "SELECT * FROM tickets";
        $stmt = $conn->prepare(query: $query);
    }
} elseif ($role == 'technician') {
    // عرض التذاكر المخصصة للمهندسين
    $query = "SELECT * FROM tickets WHERE assigned_to = ?";
    $stmt = $conn->prepare(query: $query);
    $stmt->bind_param(types: "i", var: &$user_id);
} else {
    // عرض التذاكر الخاصة بالموظفين
    $query = "SELECT * FROM tickets WHERE created_by = ?";
    $stmt = $conn->prepare(query: $query);
    $stmt->bind_param(types: "i", var: &$user_id);
}
```

Figur5.4. 4 View Ticket code

```
<header>
<h1>الفصل ١٠</h1>
</header>

<div class="filter-section">
<h2>الفصل ١٠</h2>
<div class="filter-options">
<a href="admin_dashboard.php?status=all" class="=<?php echo ($status_filter == 'all') ? 'active' : ''; >?>">الكل</a>
<a href="admin_dashboard.php?status=pending" class="=<?php echo ($status_filter == 'pending') ? 'active' : ''; >?>">الانتظار</a>
<a href="admin_dashboard.php?status=in_progress" class="=<?php echo ($status_filter == 'in_progress') ? 'active' : ''; >?>">الجارية</a>
<a href="admin_dashboard.php?status=completed" class="=<?php echo ($status_filter == 'completed') ? 'active' : ''; >?>">مكتملة</a>
<a href="admin_dashboard.php?status=delayed" class="=<?php echo ($status_filter == 'delayed') ? 'active' : ''; >?>">متأخرة</a>
</div>
</div>

<h2>البيانات</h2>
<div class="reports-table">
<table>
<thead>
<tr>
<th>رقم</th>
<th>معلومات</th>
<th>الوقت</th>
<th>الحالة</th>
<th>العمل</th>
<th>توزيع</th>
</tr>
</thead>
</table>
</div>
```

Figure 5.4.5 Admin dashboard code 1

```
<input type="text"> ->  
we action="admin_dashboard.php" method="POST">  
<input type="hidden" name="ticket_id" value="{<?php echo $ticket['id']; ?}>">  
  
<select name="technician_required">  
  <option value=""><?php echo "<?php echo '<?php echo $ticket['assigned_to']>'>" />/option>  
  <?php  
    //  
    $technician_result = $conn->query(query: "SELECT id, username FROM users WHERE role = 'technician'");  
    if ($technician_result->num_rows > 0) {  
      while ($technician = $technician_result->fetch_assoc()) {  
        <?php  
          <option value="{<?php echo $technician['id']; ?}>"><?php echo ($ticket["assigned_to"] == $technician["id"]) ? 'selected' : '' ; ?>>  
            <?php echo $technician['username'] ; ?>  
          </option>  
        <?php  
        }  
      } else {  
        echo "<option value=''><?php echo '<?php echo $ticket['assigned_to']>'>" />/option";  
      }  
    }  
  </select>  
  
<button type="submit"><?php echo '<?php echo $ticket['assigned_to']>'>/button>
```

Figure 5.4.6 Admin dashboard code 2

```

// استعلام الحيلة
$query = "SELECT * FROM tickets WHERE assigned_to = ?";
$stmt = $conn->prepare(query: $query);
$stmt->bind_param(types: "i", var: &$user_id);
$stmt->execute();
$result = $stmt->get_result();

// ارجع حالة البلاء
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    if (isset($_POST['ticket_id']) && isset($_POST['status'])) {
        $ticket_id = $_POST['ticket_id'];
        $new_status = $_POST['status'];

        $valid_statuses = ['pending', 'in_progress', 'completed', 'delayed'];
        if (in_array($new_status, $valid_statuses)) {
            echo "تم تحديث حالة البلاء";
            exit();
        }

        // تحديث حالة البلاء
        $update_query = "UPDATE tickets SET status = ? WHERE id = ?";
        $update_stmt = $conn->prepare(query: $update_query);
        $update_stmt->bind_param(types: "si", var: &$new_status, vars: &$ticket_id);

        if ($update_stmt->execute()) {
            header(header: "Location: technician_dashboard.php");
            exit();
        } else {
            echo "فشل تحديث حالة البلاء";
        }
    }
}

// إضافة تعليق
if (isset($_POST['ticket_id']) && isset($_POST['comment'])) {
    $ticket_id = $_POST['ticket_id'];
    $comment = $_POST['comment'];

    $comment_query = "INSERT INTO comments (ticket_id, user_name, comment) VALUES (?, ?, ?)";
    $comment_stmt = $conn->prepare(query: $comment_query);
    $comment_stmt->bind_param(types: "iss", var: &$ticket_id, vars: &$user_name, $comment);

    if ($comment_stmt->execute()) {
        header(header: "Location: technician_dashboard.php");
        exit();
    }
}

```

Figure 5.4.7 Technician Dashboard code1

```

<?php class tickets {
    <h2> tickets </h2>
    <div class="reports-table">
        <table>
            <thead>
                <tr>
                    <th>id</th>
                    <th>title</th>
                    <th>description</th>
                    <th>priority</th>
                    <th>status</th>
                </tr>
            </thead>
            <tbody>
                <?php while ($ticket = $result->fetch_assoc()) {
                    <tr>
                        <td><?php echo htmlspecialchars(string: $ticket['id']); >></td>
                        <td><?php echo htmlspecialchars(string: $ticket['title']); >></td>
                        <td><?php echo htmlspecialchars(string: $ticket['description']); >></td>
                        <td><?php echo htmlspecialchars(string: $ticket['priority']); >></td>
                        <td><?php echo htmlspecialchars(string: $ticket['status']); >></td>
                    </tr>
                    <tr>
                        <td><?php if ($ticket['image_path']) {
                            " style="max-width: 100px; max-height: 100px;" />
                        <?php } else {
                            <span><?php echo $ticket['image_path']; ></span>
                        <?php } >
                    </td>
                </tr>
            </tbody>
        </table>
        <form method="POST">
            <input type="hidden" name="ticket_id" value="<?php echo $ticket['id']; >"
            <select name="status" onchange="this.form.submit()" required>
                <option value="pending"><?php echo ($ticket['status'] == 'pending') ? 'selected' : ''; ><?php echo $ticket['status']; ></option>
                <option value="in progress"><?php echo ($ticket['status'] == 'in progress') ? 'selected' : ''; ><?php echo $ticket['status']; ></option>
                <option value="completed"><?php echo ($ticket['status'] == 'completed') ? 'selected' : ''; ><?php echo $ticket['status']; ></option>
                <option value="delayed"><?php echo ($ticket['status'] == 'delayed') ? 'selected' : ''; ><?php echo $ticket['status']; ></option>
            </select>
        </form>
    </div>
}
}

```

Figure 5.4.8 Technician Dashboard code 2

5.5 System testing

TestID	User Type	Test Name	Test Case	Expected Result	State
001	All users	Log In	-The user entered his information correctly -the user entered wrong username or password	- access to his page successfully -the username or password is wrong	-Successful -Successful
002	Admin	Assign a ticket to Technician	-The admin receive the ticket and send it to Technician	-The Technician receive the Ticket and solve it	-Successful
003	Admin	View All Tickets	-See the whole Tickets and sort depending on Ticket state	-All Tickets shown to Admin	-Successful
004	Technician	View Ticket & solve it and add comment	-Receive Ticket And fix it	-State change and shown in who upload the Ticket page(Student-Employee, etc.)	-Successful
005	Employee	Uploading ticket & see last tickets	- Insert the information about the Problem and upload an image (If there) -Track ticket state	-the ticket successfully go to Supervisor -Show the state and details about the ticket	-Successful -Successful
006	Student	Uploading ticket & see last tickets	-Insert information about the Problem and upload an image (If there) -Track ticket state	-the ticket successfully go to Supervisor -Show the state and details about the ticket	-Successful -Successful

Table 5.5 System Testing

5.6 Result and Discussion

The results of the program's testing were generally good, showing stable and effective performance across various scenarios. The program was able to meet a large percentage of the predefined project requirements. Additionally, the user interfaces demonstrated smooth operation and provided a positive user experience. Some points needing improvement were identified, but they do not significantly impact overall performance. The development team is committed to making the necessary adjustments to further enhance the program. Overall, the experience was successful and optimistic regarding the project's future.

5.7 Summary

Chapter 5 discusses the implementation phase of the project, highlighting its importance in transforming designs into operational systems. Key tools like Visual Studio Code, XAMPP, PHP, SQL, HTML, and CSS were utilized to create a robust development environment. The architectural design features a three-layer structure, ensuring effective processing and secure data management. Various diagrams, including sequence and ERD diagrams, illustrate system interactions and database structure. User interfaces were tailored for different roles, enhancing user experience. Testing revealed stable performance and successful requirement fulfillment, with a commitment to address identified improvements, showcasing a positive outlook for the project's future.

Chapter 6 : Conclusion and future work

6.1 Conclusion

The "Technical Support System" project has effectively addressed the limitations of Taibah reporting process by introducing a digital platform that improves service delivery, streamlines operations, and enhances user satisfaction. This computerized system allows for easier reporting, tracking, and resolution of technical issues, all while offering real-time updates and increased transparency.

By moving to a digital model, the system empowers students, faculty, and support teams, ensuring quicker and more efficient handling of technical problems.

Key accomplishments of the project include:

- Increased Efficiency: Automated reporting and tracking have minimized delays and sped up the issue resolution process.
- Better User Experience: Real-time updates and improved transparency have led to higher satisfaction among users.
- Smarter Resource Management: The system enables data-driven decisions that help allocate technical staff where they're most needed.
- Scalable Design: The system is built with future expansion in mind, making it easy to extend its use to other areas within the university.

Overall, the project not only solves existing challenges but also establishes a solid foundation for ongoing development in technical support services.

6.2 Goals Achieved

The project successfully accomplished its core objectives:

- **Custom Technical Support System:** Developed a solution specifically tailored to the operational needs of Taibah University, replacing inefficient paper-based processes.
- **Workflow Automation:** Reduced manual administrative work, enabling technicians to focus on resolving problems more effectively.
- **Faster Response Times:** A tiered system was introduced to prioritize and address urgent issues promptly.
- **Greater Transparency:** Users can now track their requests in real time, improving trust and accountability.
- **Optimized Technician Deployment:** The system uses analytics to identify issue patterns and allocate staff more efficiently.

6.3 Limitations and Future Enhancements

Despite its success, there are several areas where the system can be improved moving forward:

Planned Future Developments:

Public Launch:

Prepare the system for a broader launch to the entire university community, including training sessions and user support materials to ensure wide adoption.

Mobile App: A mobile-friendly version of the system to allow users to report and monitor issues from their phones, increasing ease of access.

Advanced Reporting Tools: Adding predictive analytics to identify frequently occurring problems and implement preventive maintenance.

AI Integration: Introducing AI-powered chatbots to handle routine inquiries and provide immediate support, reducing the workload on human staff.

Improved Notification Features: Implementing email and SMS alerts to keep users informed about ticket updates and progress.

User Feedback Mechanism: Allowing users to rate the support they receive, offering valuable insights for continuous service improvement.

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Appendix Questionnaires

الجنس

☐ ذكر

☐ أنثى

هل انت طالب / عضو هيئة تدريس

☐ طالب

☐ عضو هيئة تدريس

الكلية:

Your answer

كيف تقيم الجودة الشاملة للدعم الفني المقدم؟

☐ ممتاز

☐ جيد جدا

☐ جيد

☐ مقبول

ما مدى سهولة الإبلاغ عن المشكلات المتعلقة بالنظام الحالي؟

- ☐ سهل جداً
- ☐ سهل
- ☐ متوسط
- ☐ صعب

هل تشعر أن النظام الحالي يسمح لك بمتابعة حالة تقاريرك بشكل فعال؟

- ☐ نعم , دائماً
- ☐ بعض الأحيان
- ☐ نادراً
- ☐ لا

ما مدى أهمية سرعة الاستجابة لطلبات الدعم الفني بالنسبة لك؟

- ☐ مهم جداً
- ☐ مهم
- ☐ متوسط الأهمية
- ☐ غير مهم

هل ترغب في تلقي إشعارات حول حالة تقاريرك المقدمة؟

☐ نعم

☐ لا

هل لديك أي اقتراحات لتحسين نظام الدعم الفني؟

Your answer